



BETTA FISH · BETTA SPLENDENS

Keep your betta the right way from day one.

A complete, fact-checked printable manual: tank size and heat, the nitrogen cycle that decides everything, feeding without overfeeding, health triage, and the routines that keep a betta thriving for its full lifespan.

INSIDE

- ✓ Every temperature in degrees Fahrenheit and Celsius, and every water reading with a target number
- ✓ A full fishless cycling walkthrough, a fish-in rescue plan and a testing schedule
- ✓ Eight health pages, from fin rot to velvet, with the treatment rules and when to call a vet
- ✓ Printable emergency card, pet-sitter sheet, power-outage plan and logs

54 pages

Beginner

Print or tablet

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Every page is written to be used, not just read once.

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START HERE

How to use this package

Three ways to get value from it, depending on where you are right now.

UNITS USED THROUGHOUT

Every temperature is given in degrees Fahrenheit (°F) with degrees Celsius (°C) beside it. Tank volumes are in gallons (gal) with liters (L) beside them, sizes in inches (in) with centimeters (cm), and heater power in watts (W). Water readings are in ppm, parts per million: the unit a test kit prints, where 1 ppm is one part of a substance in a million parts of water. A percent (%) is how much of the tank's water you change, or how much of a food is protein. Prices are in United States dollars.

If you haven't set up yet

Read pages 6 to 14 in order before buying anything, and before buying the fish. The single most important rule in this book is that the tank is cycled before a betta goes in it, and cycling takes weeks, not an afternoon. Use the Budget & Shopping List on page 38 to buy once, then start the cycle on page 9 while the fish is still in the shop.

If your betta is already set up

Go straight to the Setup Checklist and Targets on page 36 and audit what you have. If the tank has no heater, no gentle filter or no liquid test kit, fix that this week. If your betta already looks unwell, start at Red Flags on page 28, but test the water first: the answer is in the test tube far more often than it is in the medicine cabinet.

Print these first

- | | |
|--|---|
| ☐ Page 36: Setup Checklist & Targets | ☐ Page 45: Pet-Sitter Sheet |
| ☐ Page 37: Emergency & Quick Targets Card | ☐ Page 46: Owner Log |
| ☐ Page 41: Symptom Quick Reference | ☐ Page 47: Equipment, Water Change & Vet Log |
| ☐ Page 42: Daily, Weekly & Seasonal Routine | ☐ Page 48: Enrichment Checklist & Log |

WHY THIS PACKAGE EXISTS

Bettas are commonly sold as bowl fish, in cups and unheated bowls, and that picture is out of date. This package reflects current tank-size, heating, cycling and enrichment standards, cross-checked against the veterinary and scientific sources listed on page 52.

NOT A SUBSTITUTE FOR

A veterinarian who treats fish. Most betta problems are solved by correcting the water, but the serious or unclear ones need an aquatic vet, and those are fewer than vets for dogs, cats or reptiles. Find one now and write the number on the emergency card.

Quick profile

Everything at a glance before you read the detail.

Scientific name	<i>Betta splendens</i> , also called the Siamese fighting fish
Adult size	2.5 to 3 in (6 to 7.5 cm). Males carry the showier fins, which is finnage, not extra size
Captive lifespan	2 to 4 years typical, up to 5 with excellent care
Native range	Thailand, Cambodia and Vietnam: shallow rice paddies, floodplains and slow streams
Diet type	Obligate carnivore (it needs meat), a surface ambush feeder. A betta-specific pellet is the staple, never flakes (page 15)
Tank minimum	5 gal (19 L), heated, filtered, cycled and lidded. Bigger is easier to keep stable (page 6)
Temperature	76 to 82°F (24.5 to 28°C), with 78 to 80°F (25.5 to 26.5°C) the sweet spot
The one hard rule	Ammonia and nitrite read 0 ppm, always. Any other reading is an emergency (page 11)
Housing	One male per tank. Clear access to the surface is a physiological need, not a nicety
Handling difficulty	Beginner, but observation only. Never bare hands (page 20)
Conservation status	Vulnerable on the IUCN (International Union for Conservation of Nature) Red List, the global list of threatened species, for wild populations

Cost overview

Initial setup

\$150 to \$360 for the equipment and first food, plus \$5 to \$30 for the fish itself (up to about \$55 for a rare variety), so \$155 to \$390 all in. Every line is on page 38.

Ongoing

\$10 to \$25 a month for food, conditioner, filter media, test reagent refills and the heater's electricity, which is \$120 to \$300 a year.

WHERE THE MONEY ACTUALLY GOES

Not the fish, and not the food. A betta eats a few pellets a day, so the real recurring cost is the electricity to run a heater around the clock. The heater and a proper filter, which the bowl-and-fish bundle leaves out, decide whether the fish thrives.

Fun facts worth knowing

- A betta breathes air. The labyrinth organ lets it take oxygen straight from the surface, an adaptation to the shallow, sometimes low-oxygen water it comes from. It also means a betta blocked from the surface has a real problem.
- Males build bubble nests out of saliva-coated air bubbles at the surface, and captive males do it with no female present.
- A betta in its shop cup can sit in fouled water for days and still be alive: Kajimura and colleagues (2023) found them unusually tolerant of ammonia, possibly from generations of shipping in small bags. Tolerance is not health.

The species, size & lifespan

What you are buying, how big it gets, and how long it stays.

One species

The pet betta is *Betta splendens*, the Siamese fighting fish. It comes from the shallow rice paddies, floodplains and slow-moving streams of Southeast Asia, in Thailand, Cambodia and Vietnam. Low-oxygen water pushed it to evolve the labyrinth organ that lets it breathe air directly from the surface. Males are famous for flowing fins and vivid colors, and for intense aggression toward other males: the name "fighting fish" comes from organized fighting matches held in Thailand centuries ago. A betta is a solitary fish, and a male lives alone.

Color and fins

Wild *Betta splendens* are a comparatively dull olive-brown. The vivid reds, blues and iridescent colors in shops are almost entirely the result of centuries of selective breeding. Males are typically priced higher than females, mostly for their showier fins and color, not for any difference in care.

Size and lifespan

Adult size	2.5 to 3 in (6 to 7.5 cm) of body. The long, flowing fins are what make a male look larger.
Lifespan	2 to 4 years is typical in an aquarium, and up to 5 is reachable with excellent care.
Wild lifespan	Not separately documented in the wild.

A betta past its second birthday is genuinely old. Split and thinning fins on a fish that eats well, swims normally and holds its color are consistent with age: the same damage that closed over in two weeks at one year old may never fully close at three. Clean water still helps, but it stops being a cure (page 30).

How the species reached the hobby

The fish entered Western science as a diplomatic gift. In 1840 the king of Siam handed breeding stock to Theodore Edward Cantor, a Danish physician, and nine years later Cantor published the first scientific account of them under the name *Macropodus pugnax*. That name was already taken, and in 1909 the ichthyologist Charles Tate Regan corrected it to *Betta splendens*, the name still in use. The hobby spread on its own track: Pierre Carbonnier was importing and breeding them in France by 1874, Paul Matte brought the first specimens into Germany from Moscow in 1896, and importers in California appear to have received them around 1910.

A FISH THAT NEEDS MORE THAN IT IS SOLD WITH

Bettas are often sold in small cups or bottles with no filter and no heater, and they tolerate that far better than most fish would, which is how the myth of the low-maintenance bowl fish survived. The care in this package is what the same animal needs to live its full lifespan: a heated, filtered, cycled tank (page 6).

Tank size, type & where it goes

The volume decision comes first, before a single plant or heater is bought.

How big

Five gallons (19 L) is the current standard minimum, and it is a floor rather than a target. Bettas can technically survive in smaller volumes, but a bigger tank is easier to keep stable: temperature and water chemistry swing less in more water, which means fewer sudden stresses on the fish. Ten gallons (38 L) gives you more margin again, and it is the size to choose if the budget allows.

The 5 gal floor holds for two reasons: more water holds steadier, and restricted space raised hiding and inactivity in the enrichment research (page 22). Water quality is a matter of management, not of container size alone, but a small tank leaves far less room for error.

TANK	WHAT IT IS FOR
5 gal (19 L)	The minimum for one betta, heated, filtered, cycled and lidded
10 gal (38 L)	More stable water for one betta, and the minimum for a group of females (a sorority)
10 to 20 gal (38 to 76 L) or more	The floor for peaceful tankmates, and a group of six standard corydoras needs a 20 gal long tank on its own count

The lid and the surface

Use a tight-fitting lid: bettas are surprisingly good jumpers. Keep clear access to the surface under it. A betta breathes air directly through the labyrinth organ, so reaching the surface is a genuine physiological need. Keep decor and plants from blocking the way up.

What does not count as a tank

NEVER

A bowl, a vase, an unheated desk container or the cup the fish came in. The cheap bowl-and-fish bundle is not a complete or humane setup: it is missing a heater and proper filtration, both of which matter for a genuinely tropical fish. It also restricts space, which the research links directly to hiding and inactivity.

Where it goes

- Out of direct sunlight. Sunlight through a window adds heat and light that neither the heater's thermostat nor the light timer controls.
- Away from drafts.
- Away from constant foot traffic and vibration. Tapping the glass startles the fish and adds stress with no benefit.

ORDER OF BUYING

Tank first, then the heater, the filter and the test kit. The fish comes last, after the tank has cycled (page 9). Setting the tank up weeks before the fish is part of the plan, not a delay.

Temperature, heater & thermometer

A betta is a tropical fish, and the heater is the item that makes every other reading more forgiving.

Sweet spot	78 to 80°F (25.5 to 26.5°C), held steady
Tolerable range	76 to 82°F (24.5 to 28°C)
Heater power	An adjustable submersible heater of 15 to 25 W for a 5 gal tank, which is about 3 to 5 watts per gallon
Rule	Consistency matters more than hitting an exact number. Avoid rapid swings

Why a betta cannot live at room temperature

Cold water suppresses a betta's immune function and digestion, and it slows recovery from everything else in this book. Below roughly 76°F (24.5°C) the metabolism and the appetite both slow noticeably, and most home temperatures without a heater run too cold. That is why a heated tank is the single biggest difference between a fish that struggles and one that thrives, and why the heater is not decor: it is the parameter that makes every other parameter more forgiving.

Cold also slows the bacteria that clear ammonia. A swing as small as 1°C (about 2°F) can cut their ammonia processing by around 30 percent, so a cold snap or a failed heater can bring ammonia back into a tank that was reading zero.

Choosing and setting the heater

- Choose an adjustable submersible heater sized for the tank: 15 to 25 W for 5 gal (19 L). For a bigger tank, work from the same 3 to 5 watts per gallon, for example 30 to 50 W for 10 gal (38 L).
- Set it to 78 to 80°F (25.5 to 26.5°C) and let it settle before the fish arrives, so the tank is warm and steady on the day.
- Read the tank with a thermometer, and check it regularly. The thermometer, not the heater's dial, tells you what the water is really doing.
- Keep the tank out of direct sunlight and away from drafts, since sun and drafts push the temperature around in ways the heater cannot see.

When the temperature moves on purpose

Two common illnesses call for opposite moves. Ich is treated by warming the tank slowly, because the parasite completes its life cycle faster in warm water (page 31). Columnaris runs the other way: the bacterium is far more aggressive in warm water, so the move is toward the low end of the betta's range, never below about 76°F (page 32). Getting the direction wrong makes either one worse, which is why the two are never guessed between.

WARM WATER HOLDS LESS OXYGEN

In a hot room or a hot spell, and any time the tank is warmed for a treatment, dissolved oxygen drops. Add surface agitation or an air stone.

Filtration & gentle flow

A betta needs a filter that cleans the water without pushing the fish around.

Why a sponge filter

A betta's long, flowing fins catch water like a sail. A power filter's outflow that a tetra barely notices can exhaust a betta and shred its fins over time, and bettas prefer calm water. A sponge filter driven by a small air pump is the standard for this species: it provides filtration with almost no current. It needs a separate air pump, and it offers limited chemical filtration, which plenty of sponge-filter setups run without. If you use another type, choose one with an adjustable flow and set it gentle. The gentle flow is the point, not the filter's brand or size.

What a filter actually does

The part you can see is the least important. The sponge or media is a surface for bacteria to live on, and those bacteria are the real filter: they turn ammonia from waste and uneaten food into nitrite, then into far less harmful nitrate. A tank can look spotless and still be quietly poisoning the fish if that colony is missing or damaged.

MEDIA TYPE	WHAT IT DOES	HOW TO HANDLE IT
Mechanical (sponge, floss, foam)	Traps uneaten food, waste and visible debris	Rinse freely, even weekly, in old tank water
Biological (sponge, ceramic rings)	Hosts the bacteria that remove ammonia and nitrite	Barely ever. Rinse gently in old tank water only, and never replace it all at once
Chemical (activated carbon)	Removes dissolved toxins, color, odor and leftover chlorine	A consumable: replace on a schedule, and take it out while medicating (page 35)

In a sponge filter the sponge does both the mechanical and the biological job, so treat the whole sponge as the colony: gentle rinse in old tank water, only when flow drops, and never under the tap (page 14).

The number on the box

The flow rating printed on a filter box is a best case, measured in clean water with no media loaded and no height to lift the water against. Real flow is lower, especially as the media collects debris. For a betta that is a reason to check the surface, not to buy a stronger filter: what matters is that the water is moving enough to bring oxygen in and gentle enough that the fish can rest.

WHAT HAPPENS WHEN A FILTER STOPS

The colony needs continuous, oxygenated flow. Nitrifying bacteria become inefficient once dissolved oxygen in the filter drops below roughly 2 ppm, and biological filters do not take rapid change well. A pause while you clean is harmless. A longer stop is an emergency with its own steps (page 43).

WHERE IT GOES

Put the sponge filter in the tank before the cycle starts, and leave it running. A filter that has never run has no bacteria in it yet.

The nitrogen cycle & fishless cycling

Weeks of work with no fish in the tank. This is the part that gets skipped, and it decides everything.

Fish waste and uneaten food break down into ammonia, which is toxic even in small amounts. One group of bacteria converts ammonia into nitrite, which is also toxic. A second group converts nitrite into nitrate, which is far less harmful and is what routine water changes remove. "Cycling" a tank is growing both groups until they keep pace with the ammonia the fish produces every day.

STAGE	WHAT IS HAPPENING	WHAT A TEST SHOWS
1. Ammonia	Waste and leftover food break down into ammonia	Ammonia climbs, nitrite and nitrate at 0
2. Nitrite	Bacteria convert ammonia to nitrite, which blocks oxygen uptake in the blood	Ammonia falls, nitrite spikes
3. Nitrate	A second group converts nitrite to nitrate	Nitrite falls to 0, nitrate appears
4. Cycled	Both colonies keep pace with the daily waste	Ammonia 0, nitrite 0, nitrate present

Fishless cycling, step by step

1. Set the tank up completely, filter running and heater at its normal setting. Warm, stable water cycles faster, and cold slows the bacteria.
2. Fill with tap water dosed with a conditioner that handles both chlorine and chloramine. Either one kills the bacteria you are trying to grow.
3. Dose pure ammonia, with no surfactants or dyes, to 3 ppm, inside the usual 2 to 4 ppm range. Food also works as an ammonia source, but it rots and fouls the water while you wait.
4. Seed the tank if you can: filter media, gravel or sponge from an established, healthy tank carries a living colony and is the most reliable way to shorten the process. Bottled bacteria have a mixed record, so treat any single product's claims with some skepticism.
5. Test daily or every other day, and dose ammonia back up to 3 ppm whenever it drops, so the bacteria stay fed. Skip water changes during the cycle, because they remove the ammonia you are dosing.
6. The cycle is done when a full ammonia dose reads zero within 24 hours, nitrite also reads zero, and nitrate has started to build. All three must be true before a fish goes in.
7. Then change enough water to bring nitrate under 20 ppm (page 12), keep the heater running, and bring the fish home.

HOW LONG

Most fishless cycles take four to six weeks at a warm, stable temperature. In real home aquarium filters, two tanks reached undetectable ammonia and nitrite by about week three and a third took until week eight. Cold water, no seeding and an inconsistent ammonia dose all push a cycle toward the slow end. There is no way to rush the biology, only ways to seed it. More ammonia is not faster: high ammonia holds back the group that converts nitrite, so stay at 3 ppm.

Fish-in cycling & when a cycle stalls

The rescue plan for a fish that came home before the tank was ready, and the checklist for a tank that was fine and no longer is.

If the fish is already home and the tank is not cycled

This is a fish-in cycle. It is the riskier path, and it is a rescue, not a plan, but it is manageable with attention every day.

- **Test daily.** Ammonia and nitrite are what you are watching; nitrate matters far less at this stage. Use a liquid reagent test kit.
- **Act on a rising reading, not on a crisis.** The action line is 0.1 ppm of ammonia or nitrite, which calls for a water change of at least a third of the tank. A reading that is clearly climbing toward toxic calls for 50 percent or more, done without hesitation, because a large change that interrupts the cycle is a far smaller risk than leaving the fish in rising ammonia. Day to day, most fish-in cycles are managed in the 0.1 to 0.25 ppm range, with the size of the change scaling up as the reading gets worse.
- **Use a conditioner that detoxifies ammonia** between changes. It buys a margin of safety without killing the bacteria you are growing, but it is a bridge, not a substitute for testing and water changes.
- **Never rinse filter media under the tap.** Tap water carries chlorine or chloramine because they kill microbes, and the media is covered in the microbes you are trying to keep alive.

The new tank problem

In the first two to three weeks of a new tank, ammonia rises because the filter's bacteria have not finished establishing, and overfeeding and overstocking make it worse. If a fish in a tank running less than a month suddenly looks sick, the nitrogen cycle is the first suspect, not disease.

Why a cycle stalls or restarts

A tank that was reading zero can suddenly show ammonia or nitrite again. The usual causes are all avoidable:

- **Overdosing ammonia**, which inhibits the nitrite-converting group.
- **A crash in pH or in KH.** pH is how acidic the water is, and KH (carbonate hardness) is the buffer that holds it steady. The bacteria need both to stay stable.
- **Untreated tap water.** Chlorine or chloramine kills bacteria the way it kills pathogens.
- **Medicating the tank.** Several common medications, including antibiotics, kill filter bacteria, and copper is harder still: ammonia and nitrite can stay elevated for weeks to months afterward (page 35).
- **Rinsing media under the tap**, or replacing all of it at once instead of a portion at a time.
- **A temperature drop**, since the bacteria are measurably slower in cold water.

WORK BACKWARD

If a cycled tank reads ammonia or nitrite again, run down that list before assuming something is wrong with the fish. It is fixable the same way an initial cycle is managed: test, change water, and give the bacteria time to recover.

Water quality, testing & hardness

Most betta illness is a water problem wearing a costume. These are the numbers behind that.

READING	TARGET	WHY IT MATTERS
Temperature	78 to 80°F (25.5 to 26.5°C)	Cold water suppresses immune function and digestion
pH	6.5 to 7.5	Wide tolerance. Stability matters more than the exact number
Ammonia	0 ppm	Even 0.25 ppm causes gill damage. Any detectable reading is an emergency
Nitrite	0 ppm	Blocks oxygen uptake in the blood. Toxic at any measurable level
Nitrate	Under 20 ppm	The byproduct of a healthy cycle, controlled by regular water changes

Ammonia and nitrite are the only two readings with no safe nonzero baseline: a cycled tank reads exactly 0 ppm for both. A kit reads total ammonia, and a higher pH pushes more of it toward the toxic form, so the same reading can be more dangerous in one tank than another.

Test with a liquid kit, not strips

A liquid reagent kit that reads ammonia, nitrite, nitrate and pH is the only reliable way to know whether a tank is cycled. Paper strips are notoriously imprecise at the low end of the ammonia and nitrite scales, exactly where the answer matters.

WHEN	WHAT TO TEST
New or recently set up tank	Ammonia and nitrite every 1 to 2 days until both read zero for at least a week
Established, cycled tank	Weekly at minimum. Established tanks can still spike if the filter is disrupted, the media is rinsed in tap water, or the tank is overfed
The betta seems off	Immediately. Clamped fins, lethargy and lost appetite often show in the water before they show on the fish
After a water change over 30%	A quick recheck confirms the new water did not bring chlorine or chloramine in

Hardness: GH and KH

General hardness (GH) is the amount of dissolved calcium and magnesium, a mineral supply that does not buffer anything. Carbonate hardness (KH) is the water's buffering capacity: its ability to neutralize incoming acid and hold pH still. Every tank makes acid, and the bacteria use up carbonate as they work, so a tank can hold a normal pH for weeks on a borrowed buffer and then drop hard when it runs out. Bettas tolerate a fairly wide range of both. The risk is very soft, poorly buffered water, common with reverse-osmosis or heavily filtered tap water, because pH can crash after a water change, which is more stressful than a steady pH slightly off target. Test KH before reaching for any pH adjuster. Bottled pH-up and pH-down force the pH away from its buffered baseline without fixing the cause, and can overshoot badly. Straight reverse-osmosis water has almost no buffer, so it is blended with tap water. If KH stays very low and pH swings, a small, consistent dose of a buffering product resolves it, after ammonia, nitrite and temperature are right.

Water changes & conditioner

The single most useful habit in fishkeeping, and the easiest one to do in a way that undoes itself.

What a change removes that a filter cannot

The filter's bacteria turn ammonia into nitrite and nitrite into nitrate, and then stop. Nothing in an ordinary freshwater filter removes nitrate, so it accumulates until water carrying it leaves the tank. A water change is not cleaning in the sense of removing visible dirt: it is dilution of something invisible. A clear tank is not a clean tank.

The number, and the test that moves it

For a heated, filtered betta tank the baseline is 25 to 30 percent of the water once a week. In a 5 gal tank that is about 1.25 to 1.5 gal (4.7 to 5.7 L). Small tanks concentrate waste faster than the volume suggests, and a smaller tank can need more frequent partial changes. Test nitrate before the weekly change and let the reading set the size:

NITRATE BEFORE THE CHANGE	WHAT IT MEANS	WHAT TO DO
Under 20 ppm	The schedule matches the load	Keep the 25 to 30 percent weekly change
20 to 40 ppm	Nitrate is winning slowly	Move to 40 to 50 percent weekly, or add a midweek change, and feed less
Over 40 ppm	Past the safe line	A 50 percent change now with matched water, retest in two days, find the cause

Small changes lose the arithmetic: a tank that adds 20 ppm of nitrate a week and gets a 25 percent change settles at about 80 ppm, while a 50 percent change settles near 40.

Putting the new water in

- Chlorine first.** Tap water carries chlorine or chloramine, and chlorine damages gills within minutes with no treatment afterward. Chloramine is chlorine bonded to ammonia, so use a conditioner labeled for both. Dose it as the water goes in, a third before the water starts, a third at the halfway mark and a third at the end.
- Temperature next.** Mix hot and cold to land within a couple of degrees of the tank, checked with the same thermometer you read the tank with. A change that drops the tank several degrees is the classic trigger for a white-spot outbreak a week later.
- Siphon from the substrate, not the surface.** A gravel vacuum pulls out the waste that settles into the gravel, which is where most of the nitrate is made. Work over a different section each week.
- Rinse the filter sponge in the bucket of old tank water** while it is out, never under the tap.

DO NOT DO THE BIG CATCH-UP CHANGE

A tank that has gone months without a change has drifted far from the tap. Swapping most of the water at once shocks fish that were coping. That tank gets small daily changes instead: 10 to 15 percent a day for the first week, every other day in the second, then 20 to 25 percent twice a week for two weeks, testing as you go. A new tank that is still cycling follows the fish-in rules (page 10).

Substrate, plants, light & decor

The inside of the tank, and the one fact about it that surprises people: cover makes a betta swim more, not less.

Substrate

Choose a soft substrate: sand or smooth gravel. Nothing sharp, since a betta's fins tear on anything rough. The gravel vacuum pulls waste out of it at each water change.

Plants and hides

Provide hides and plants. Live plants are the better choice: fish in aquaria planted with *Sagittaria subulata* showed the longest average active swimming time and fewer negative behaviors, and live planting also affects water quality and the tank's microbial community. If you use artificial plants, choose soft silk ones. Stiff plastic tears betta fins, a real and common injury and an open door for fin rot (page 30). Add something near the surface for resting, such as a leaf hammock or a broad-leaved plant. Bettas are labyrinth fish and surface breathers, so resting places near the top matter as much as cover below. Do not add so much decor that the fish cannot reach the surface, because air access outranks everything else on this page. Driftwood and decor placed to break the tank into zones give a betta somewhere to be that is not simply the middle of a glass box. Driftwood also releases tannins, which tint the water faintly amber and soften the light. That is closer to the blackwater conditions the species comes from, and it is cosmetic.

Light

A betta needs no special light, but it needs a day and a night. Put the light on an outlet timer so it switches on and off at the same time every day.

SITUATION	HOURS OF LIGHT A DAY
A newly planted tank	6 to 8, while the plants settle
Once the plants have grown	8 to 12
Algae taking over	Shorten the day again

Keep nights dark. Light left on around the clock does not help the plants, which still need their night, and it hands the extra light to algae. In a 2012 breeding study, Giannecchini, Massago and Fernandes kept pairs under five schedules: pairs on 12 or 16 hours of light spawned most often and laid the most eggs, while pairs under 20 hours or constant light laid far fewer. For a pet betta, the working rule is a normal-length day, a dark night and no light left on overnight. A betta lit hard from above in a bare tank has nowhere to be out of the glare.

THE RESEARCH BEHIND IT

Restricted, bare conditions raised hiding and inactivity, while planted tanks produced the most active swimming. Cover makes a betta more willing to use open water, because being exposed is only a risk when there is nowhere else to be (page 22).

Cleaning & hygiene

Every other cleaning step in fishkeeping is forgiving. Filter care is not.

The one rule that matters

NEVER UNDER THE TAP

Rinse biological media in a bucket of water siphoned out of the tank during a water change, swish it, and put it straight back. Tap water carries chlorine or chloramine, added to kill bacteria, and the media is covered in the bacteria colony the tank depends on. A few seconds of tap water on a sponge is a small-scale version of a disinfection pass that kills the colony and sends the tank back to day one.

What to clean, and how

WHAT	HOW
Sponge filter	Gentle rinse in old tank water, only when flow drops. Never replace the whole sponge at once: swap a portion at a time, weeks apart, so the untouched media keeps the colony going while the new media is colonized
Substrate	Vacuum a different section at each water change rather than the whole floor at once
Uneaten food	Remove anything left after 1 to 2 minutes. Every pellet that reaches the gravel becomes ammonia
Activated carbon, if used	Replace on a schedule, and out of the filter while medicating

Keep equipment from tank to tank

- A quarantine or hospital tank keeps its own nets and siphon hoses, and they never touch another tank. A net that moves between the quarantine tank and the main tank undoes the separation the second tank exists for.
- Between uses, disinfect the hospital tank and its equipment and store it dry.
- Columnaris is highly contagious between tanks on shared equipment, so nets, siphons and buckets are disinfected before they touch another tank (page 32).

Hands, gloves and the tank

Fish tuberculosis is rare but it can pass to people through aquarium water, especially through broken skin. Use gloves if you have any open cut and are doing maintenance on a tank you suspect is affected (page 33).

AFTER MEDICATING

A treatment can stall the bacteria and cause an ammonia spike on top of the illness, and copper keeps nitrite and ammonia elevated for weeks to months. Keep testing ammonia and nitrite long after the course itself has finished.

Diet & feeding schedule

Overfeeding is the single most common mistake in betta keeping, and it fouls the water too.

What a betta eats

Bettas are obligate carnivores and surface ambush feeders: in the wild they snatch insects, larvae and small crustaceans off the water's surface. In a tank the staple is a betta-specific, high-protein pellet. Tropical flakes and goldfish pellets are the wrong food, with the wrong nutrient profile and more filler.

How much: portion matters more than frequency

A betta's stomach is roughly the size of its own eye, which is the real basis for the standard guidance of 2 to 4 pellets per meal. Depending on pellet size and brand that can run from 2 or 3 up to 6 or 7, so the more reliable rule is whatever the fish clears within 1 to 2 minutes, with nothing left sitting on the substrate afterward. Uneaten food is the fastest way to foul a small tank.

How often

Once or twice a day is the usual range, and up to three smaller meals spaced 6 to 8 hours apart also works. There is no single correct number: pick a routine you can hold consistently, and watch the fish's body condition rather than chasing an exact figure (page 19). Build in one fasting day a week, with no food at all, to support digestion and reduce the risk of bloating and constipation.

PRE-SOAK EXPANDING PELLETS
If a pellet expands in water, soak it first to reduce the risk of bloat. Freeze-dried foods get the same treatment: rehydrate them before feeding, because fed dry they raise the constipation risk.

By life stage

STAGE	FEEDING
Fry, freshly hatched	Infusoria (a mixture of very small aquatic organisms) several times a day
Fry, free-swimming	4 to 5 small meals a day
Weeks 3 to 4	3 to 4 small meals a day
Juvenile	2 to 3 meals a day, moving onto the adult routine once separated into its own space
Adult	Once or twice a day, 2 to 4 pellets a meal, one fasting day a week

A WEEKEND AWAY IS LOW RISK
A short absence, such as a weekend with an automatic feeder, is low risk for a healthy adult. The point to investigate a refused meal is days, not hours (page 18).

Pellets, protein foods & the food chart

Staple every day, occasional a few times a week, limited, and never at all.

Reading a pellet label

Look for a betta-specific pellet that lists dried meats such as brine shrimp, krill or fish meal near the top of the ingredient list, and roughly 30 percent protein or higher. A good range for a betta is about 35 percent protein and about 5 percent fat. Not tropical flakes, and not goldfish pellets.

Food chart

FOOD	TIER	HOW TO OFFER IT
Betta-specific pellet	STAPLE	The backbone of the diet: 2 to 4 pellets a meal, pre-soaked if it expands
Frozen, freeze-dried or live brine shrimp	OCCASIONAL	1 to 3 times a week. Freeze-dried food is rehydrated first
Daphnia	OCCASIONAL	1 to 3 times a week
Mysis shrimp	OCCASIONAL	1 to 3 times a week
Mosquito larvae	OCCASIONAL	1 to 3 times a week
Tubifex worms	OCCASIONAL	1 to 3 times a week
Bloodworms	LIMIT	1 to 2 times a week at most, just 1 to 2 worms a meal. They are rich, and bettas will gorge on them given the chance
A blanched, shelled pea	REMEDY	One small piece now and then as a natural remedy for constipation, not a regular food
Tropical flakes, goldfish food	NEVER	The wrong nutrient profile, with more filler, as a main diet

Variety is the cheapest enrichment

Round out the pellet with frozen, freeze-dried or live protein foods 1 to 3 times a week. Move the feeding point around the tank rather than dropping food in the same corner every day, and vary the food itself. It is the cheapest item on the enrichment list, and the one most consistent with the wider fish enrichment research, where structural and feeding variation both show effects (page 22).

Treats & the never-feed list

What is safe in moderation, and what does harm.

Safe, in moderation

Frozen or freeze-dried bloodworms, brine shrimp, daphnia, tubifex worms, mysis shrimp and mosquito larvae, and a blanched, shelled pea now and then as a remedy for constipation. All of them sit on top of the pellet staple and never replace it.

Never feed

FOOD	WHY NOT
Bread	Swells and causes bloating, with little nutritional value
Onion and garlic	Toxic sulfur compounds
Salt, and sugary foods	Salt is a water treatment for scaled fish, never a food (page 35), and sugar has no place in a carnivore's diet
Processed meats, such as hot dogs or lunch meat	Never appropriate for a betta
Citrus	Acidic enough to crash the tank's pH and burn the gills
Dairy	Bettas are effectively lactose intolerant
Tropical flakes and goldfish pellets as a main diet	The wrong nutrient profile, and more filler

Overfeeding, and what it looks like

Overfeeding leads to bloating, constipation, swim bladder trouble and fouled water from what is left uneaten (page 33). An overfed fish shows a swollen belly and may pass stringy feces or none. The remedy is smaller portions, the fasting day, and a look at the fish from above every week (page 19), not a different food.

IF A PELLET SWELLS IN WATER, IT SWELLS IN THE FISH

Soak expanding pellets before feeding, and rehydrate anything freeze-dried. This one habit prevents a large share of bloat.

A FISH THAT LOOKS HUNGRY IS USUALLY A FISH THAT ALWAYS LOOKS HUNGRY

A betta's stomach is about the size of its own eye, so a few pellets fill it. A betta that begs at the glass is not a betta that needs more.

Why a betta stops eating

Bettas refuse food for ordinary reasons and for one worrying one. Work them in order.

The six reasons

1. **Environmental stress.** A new tank, a recent move, handling or an aggressive tankmate can suppress appetite temporarily.
2. **Poor water quality.** An ammonia, nitrite or nitrate spike is one of the most common causes of both appetite loss and general illness, and it is the first thing to check, every time.
3. **Water that is too cold.** A betta needs roughly 76 to 82°F (24.5 to 28°C). Cold water slows metabolism and appetite, and most homes without a heater run too cold.
4. **Picky eating.** Bettas are known for rejecting a new brand or a new type of food outright. Try a different food before assuming illness.
5. **Overfeeding, constipation or swim bladder disease.** Eating too fast or too much can swell the stomach against the swim bladder. Watch for a bloated belly, stringy feces, or floating and sinking unnaturally.
6. **Illness.** Bacterial infection or parasites can suppress appetite, often alongside white spots, clamped fins or labored swimming.

What to try, in order

STEP	
1	Test ammonia, nitrite and nitrate, and read the thermometer. Do this before anything else
2	Fix whatever the test found. Correcting the water is the treatment, not a delay
3	Look the fish over: swollen or bloated belly, stringy or absent feces, lethargy or hiding, clamped fins, white spots, a curved or S-shaped spine, floating or sinking, labored swimming
4	Offer something different. A new brand or a different protein food is the usual cure for a picky eater
5	Check what changed around the tank, and give a new tank or a move a few days to settle

When a fast becomes a symptom

THE NUMBERS

A healthy adult can physically survive up to about two weeks without food, but that is an emergency ceiling, not a target. Start investigating at 5 to 7 days of refused food, after you have already checked water quality and temperature and tried a different food. Start sooner, at once, if any physical symptom from step 3 shows up alongside it. A week or more of refusal plus a physical symptom is the real concern threshold.

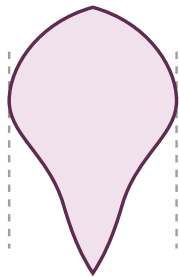
A short absence of a day or two, a weekend away, is low risk. Most refusals come down to stress from a move or a tank change, pickiness about a new food, or water that is too cold or of poor quality, which is why those are the first things to check. Bring the Owner Log to any vet call (page 46).

Body condition & portion control

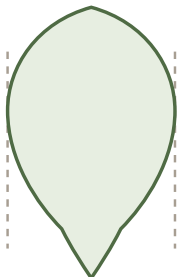
You cannot weigh a betta at home. You can read its shape from above.

The top-down check

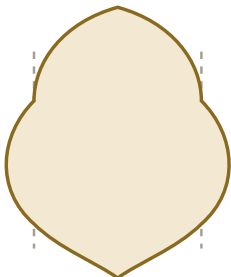
Look down into the tank from the top, not through the side. A betta in good condition looks like a streamlined torpedo, with the "shoulders" just behind the gill covers the same width as the middle third of the body. If the sides bulge outside the shoulders, the betta is too fat. If the shoulders stand out and the middle is small, it is too thin.



Too thin



Correct



Overweight

The dashed guides mark the shoulder width. The middle third should reach them and go no further.

SHAPE	WHAT YOU SEE	WHAT TO DO
Too thin	Prominent shoulders and a small middle	Work through the refusal checklist
Correct	Shoulders and the middle third the same width	Hold the portion. Look again each week
Overweight	Sides bulge outside the shoulders	Cut each meal by one pellet, keep that for 2 weeks and look again. If nothing has changed, take away another pellet

Setting the portion

Begin with the number of pellets that could, in theory, fit inside the fish's eye, which is the basis of the 2 to 4 pellets a meal in the diet section. Keep the weekly fasting day. A good staple pellet runs about 35 percent protein and about 5 percent fat (page 16).

WHY IT MATTERS

Overfeeding leads to bloating, constipation, swim bladder trouble and fouled water. A weekly top-down look catches the trend long before the belly is swollen, and it is the closest thing a betta owner has to a weigh-in. Write the result in the Owner Log (page 46).

Handling: net, cup & never bare hands

A betta is not a hands-on pet. You will almost never handle this fish, so reading it from outside the glass is the whole skill.

Why bare hands are off limits

A betta's slime coat is a genuine barrier against infection and parasites. It works around the clock, and a bare hand strips it away on contact. Every time a betta is touched directly, that protection takes a hit, and a technique that seems harmless, such as briefly cupping the fish in your hands, does real damage even when nothing looks wrong afterward. Move the fish with a net or a cup, never scoop it up by hand.

NEVER

Pick a betta up by hand, and never pour the water from the shop's cup or bag into your tank. Float the sealed cup, then net the fish out.

What handling looks like day to day

- Net or cup transfers only, for moving to a new tank or the hospital tank, or when a water change forces it.
- Avoid tapping the glass. It startles the fish and adds stress with no benefit.
- Keep the tank away from constant foot traffic and vibration if you can.
- Beyond that it is observation, not direct interaction. The relationship with a betta is built through good water quality and careful watching, not touch.

Reading your betta instead of touching it

A HEALTHY BETTA

Good, vibrant color. Intact, flowing fins carried open. Normal swimming activity through the tank, not one corner. Eager interest in food. Regular trips to the surface for air, then back down.

STRESS SIGNS

Clamped fins pressed tight to the body. Faded color. Lethargy and hiding. Rapid gill movement. Skipped meals. Almost all of it traces to water quality or a temperature that has drifted, not to anything done by hand.

TEST THE WATER FIRST

When a betta looks off, test the water before you reach for a diagnosis, and well before medication. A large share of what looks like disease is a water quality problem (page 28).

MOVING IT WHEN YOU MUST

A net or a cup, one fish at a time, and into water at the same temperature. The hospital tank and the main tank each keep their own net (page 14).

Body language, flaring & bubble nests

The behavior problems in this species are mostly housing problems, and they show on the fish before they show in the water.

Reading the fish

BEHAVIOR	WHAT IT USUALLY MEANS	FIRST THING TO CHECK
Clamped fins	A posture, not damage: the fins are held shut against the body. Distress, and one of the more universal signs of stress or illness	The water, then the temperature
Faded or washed-out color	A general illness indicator that often shows before anything more specific	The water, temperature, and anything new nearby
Resting on a leaf or the bottom, especially after lights out	Often just sleeping. Learn your own fish's normal resting posture so you can tell the difference	Whether it is new, constant, or comes with other signs
Gulping at the surface	Normal: that is the labyrinth organ. Constant, urgent gulping is not, and is more often an oxygen or ammonia problem than a disease	Ammonia and nitrite, and surface movement
Hanging in the current by the filter	Often chasing the most oxygen-rich water in the tank	Test the water
Flashing: rubbing against decor	An irritant on the skin or gills, commonly a parasite, sometimes water irritation	Test, then look closely for white grains or gold dust
Hiding constantly in a bare tank	The restricted-space pattern from the research	Tank volume, planting and cover
Building a bubble nest	A male doing what males do, with or without a female	Nothing, this is the good one

Flaring, and what to do about it

Flaring is the threat display: gill covers spread, fins fanned. A male does it almost at once when he meets another male, and females flare too, just less dramatically. It is normal behavior, and a fish that never flares is not necessarily happier. Flared gill covers and spread fins aimed at a newcomer are a display. A display that turns into chasing and torn fins, on either fish, means the pairing has failed and one fish needs another home (page 26).

A MIRROR IS A STRESS TEST, NOT A TOY

A mirror provokes a threat display, and the stress hormone cortisol rises with it. Brief exposure is sometimes suggested as exercise, but what you are doing is triggering a threat display, so it belongs at the bottom of the enrichment list or off it entirely.

The order to check things in

Whatever the behavior, the sequence is the same: water first, then temperature, then anything that changed in or around the tank, then the fish itself. Working in that order stops you treating a fish for an illness it does not have.

Enrichment: what the research says

The betta is the most-studied fish in the freshwater hobby, and the results are inconvenient for how it is usually sold.

What the research found

Space and environmental enrichment both affect behavior in *Betta splendens*, with restricted space raising signs of stress including hiding and inactivity. Tank size and furnishings change behavior and welfare too. Planting matters as well: fish in aquaria planted with *Sagittaria subulata* showed the longest average active swimming time and fewer negative behaviors.

THE COUNTERINTUITIVE PART

Adding dense cover does not produce a fish you never see. It produces one that swims more, because being out in open water is only a risk when there is nowhere else to be. Live planting also affects water quality and the tank's microbial community, so it works on two axes at once.

Priority order

1. Tank volume, decided before anything goes in it. Larger tanks promoted natural, active behavior, and restricted conditions produced the hiding and inactivity that gets misread as a calm fish.
2. Dense planting and cover, live where you can, and soft artificial plants if not.
3. Surface resting spots and broken sightlines: broad-leaved plants, a leaf hammock, and driftwood placed to divide the tank into zones.
4. Varied feeding location and varied food.
5. Tannins from driftwood and softened light.
6. Mirrors, at the bottom of the list or off it entirely.

On group housing

Bettas of either sex can be group-housed with lower aggression if they have been housed in groups since hatching under enriched conditions; when a young fish is taken out of an enriched setting also shapes its later aggression and sexual maturity. That is a finding about rearing, not a license to combine adult males: two adult males bought separately are still two adult males.

Don't read a still fish as a calm one

A betta sitting motionless in a bare tank is displaying the pattern the restricted-space study produced. Stillness in an empty box is not contentment, and it is the easiest welfare problem in this species to miss.

WHAT NOT TO DO

Do not use a mirror as a daily routine. Do not use stiff plastic plants, which damage fins. Do not combine adult males on the strength of the group-housing research. Do not treat a heater and filter as optional extras, since no amount of enrichment compensates for unstable water.

Common mistakes & myths

The mistakes that cost bettas their lives, and the myths that keep them going.

MISTAKE	WHAT ACTUALLY HAPPENS
Adding the fish to an uncycled tank	Ammonia has nowhere to go. This is why bettas frequently decline in the first two weeks of a new, clean-looking tank (page 9)
No heater	Cold water suppresses immune function and digestion, and slows recovery from everything else
A bowl, a vase, or the cup it came in	No filtration, no steady temperature, and restricted space, which the research links to hiding and inactivity
Overfeeding	Bloating, constipation, swim bladder trouble, and fouled water from what is left uneaten
Strong filter flow	A betta's fins catch the current like a sail. Constant swimming against it exhausts the fish and can shred its fins over time
Stiff plastic plants and sharp decor	Torn fins, a real and common injury and an open door for fin rot
Testing with paper strips, or not at all	Strips are imprecise exactly where precision matters, at the low end of ammonia and nitrite
Reaching for medication first	Medicating before ruling out the water can add stress to a fish already struggling with its environment, and a medicated tank can lose its bacteria colony
Using a mirror as a daily toy	It triggers a threat display and a measurable stress response

Myths worth retiring

THE MYTH	WHAT IS ACTUALLY TRUE
"A betta is happy in a small bowl"	Restricted space raised hiding and inactivity in the research. Nothing in it shows a preference for less room
"It breathes air, so it does not need a filter"	The labyrinth organ solves oxygen. Waste still builds up as ammonia, which is what the filter's bacteria handle
"A bowl-and-plant kit is a complete setup"	It is missing a heater and proper filtration, the two things that matter most for a tropical fish
"A cup betta that looks fine is healthy"	Bettas tolerate fouled cup water for days. Whether one is healthy is a question for the fish itself, not the water
"Flaring at a mirror is good exercise"	It is a threat display, and it raises the stress hormone cortisol
"A still betta in a bare tank is relaxed"	That is the pattern the restricted-space study produced

IF YOU FIX ONLY THREE THINGS

Cycle the tank before the fish goes in, add a heater and a gentle filter, and use a lid. Most of the rest of this book is detail on those.

Choosing a healthy betta & bringing it home

The price tag says nothing about the fish's condition, and the cup it waits in says less than it seems to.

Judging the fish, not the water

Bettas are sold in small cups or bottles without filtration or aeration, and they tolerate that better than most fish would. A 2023 study by Kajimura and colleagues put the 96-hour lethal level of the toxic form of ammonia for bettas at 7.7 milligrams per liter, and the authors suggest generations of shipping in small bags that hold about 90 percent air and a little water may have selected for that tolerance. A betta can sit in fouled water for days and still be alive, so the cup tells you little. Whether the fish is healthy is a question for the fish itself.

WHAT A HEALTHY BETTA SHOWS

Bright, vibrant color. Intact fins, held open. Normal activity, swimming around the cup and not hiding or lying on the bottom all the time. A good appetite, if the store will let you watch it eat.

WHAT TO WALK AWAY FROM

A fish that is not eating, or is lethargic or swimming abnormally. Frayed fins, or fins that look worse on a second visit. New spots or growths, including the gold dust of velvet or the salt-grain spots of ich. Fins pressed flat against the body.

A sick \$5 fish is the most expensive betta in the store once the medication and a hospital tank are added to the bill. Paying a few dollars more for a lively fish in cleaner water is the cheapest insurance in betta keeping.

Before it comes home

- The tank is cycled: a full ammonia dose reads zero within 24 hours, nitrite reads zero and nitrate is present (page 9).
- The heater has been running long enough to hold 78 to 80°F (25.5 to 26.5°C) steadily.
- You have found a veterinarian who treats fish. The American Association of Fish Veterinarians keeps a find-a-vet directory that covers North America, and if it shows no one within reach, ask a local vet to take the case with a fish vet's guidance (page 29).

The trip and the introduction

1. Float the sealed cup or bag on the tank's surface for 20 to 30 minutes so the water inside comes to the tank's temperature.
2. Net the fish out and release only the fish. The cup water carries the shop's ammonia and whatever its tanks were carrying, so it never goes into your tank.
3. When the cup and the tank differ by more than 10°F (5.5°C), or the pH and hardness are very different, acclimate slowly by adding tank water to the cup a little at a time instead. A gap past about 18°F (10°C) stretches that to two hours or more.
4. Add one fish at a time, and watch the water parameters for a day or two after any addition.

THE FIRST DAYS

Stress from a new tank or a move can suppress appetite temporarily. A refused meal in the first days is an expected stress response, not yet a symptom, and the five-to-seven-day investigation line on the feeding pages applies after the water and temperature have been checked (page 18).

Quarantine & the hospital tank

A fish that looks fine on arrival is the normal state of a fish whose symptoms have not shown yet.

Why every new fish is quarantined

Quarantine means keeping a new or sick fish in a separate aquarium for a set time before it joins others. A pathogen picked up somewhere between breeder, wholesaler and shop does not always show on day one, and by the time it does in a shared tank, your other fish have already been exposed. The isolation also makes the fish easy to watch and to treat.

HOW LONG

At least 30 days. For a fish you especially do not want to lose, or do not want introducing something to an established tank, 30 to 60 days. Thirty days is the floor, not a target to hit and stop, because some parasites and infections take that long to show.

The bare hospital tank

"Bare" matters: no substrate, no driftwood, nothing with crevices a parasite or its eggs can hide in, and nothing that makes the tank harder to see into or scrub clean between uses. The home version is cheap.

Tank	10 to 20 gal (38 to 76 L), kept bare, sized to what you are quarantining
Filter	A sponge filter on an air pump: gentle biological filtration, and no cartridge to replace mid-quarantine
Heater	Matched to the species and the treatment. For a betta, its normal range
Air pump	Runs the sponge filter and keeps oxygen up during any treatment
Test kit	A hospital tank's filter is smaller and more fragile than the main tank's, so ammonia and nitrite checks matter more

A brand-new sponge has no bacteria in it. Seed the hospital tank's sponge from an established, running filter: keep a spare sponge cycling permanently in the main tank's filter, or borrow one for a few days before you need it, so the filter works on day one and you are not running a fish-in cycle in a tank you are also trying to treat (page 10).

What to watch for in quarantine

Feed normally, do routine partial water changes, and check daily for anything that was not there on arrival: clamped fins, unusual spots or patches, rapid gill movement, flashing against decor, appetite loss, cloudy eyes, or a fish that hangs listlessly. A fish that arrives thin or stressed can take a few days to settle before its true baseline shows, so do not read the first day or two too literally in either direction.

Keep it separate

- The quarantine tank keeps its own nets and siphon hoses, which never touch another tank.
- Between uses, the tank and equipment are disinfected and stored dry. A tank that still has residue from the last sick fish is not starting the next 30 days from zero.
- Treat in the hospital tank, not the display. Several medications kill the bacteria that run the filter, and copper is extremely toxic to invertebrates and plants (page 35).

Tankmates

A betta does not need company. Anything you add is a risk you choose on the fish's behalf.

The fighting-fish name is earned between males. A male lives alone or with tankmates that are peaceful, short-finned and not interested in his fins. The tank also has to be bigger than a lone betta's: plan on at least 10 to 20 gal (38 to 76 L) for peaceful nano fish.

OPTION	VERDICT	NOTES
Alone, in a planted 5 to 10 gal tank	DEFAULT	The setup this whole package is written around
Corydoras catfish	POSSIBLE	Peaceful and short-finned. A group of six standard corydoras needs a 20 gal long tank on its own count
Mystery snails	POSSIBLE	Peaceful, and leaves the betta's fins alone
Ghost shrimp	RISKY	10 gal or more, with plenty of hiding places, and only shrimp that have passed about 1.5 in (4 cm). Smaller or juvenile shrimp get eaten. Adding the shrimp before the betta helps
Some small tetras or rasboras	RISKY	Peaceful, short-finned, warm-water species only. A fast school such as zebra danios stresses a slow, long-finned betta even without real aggression
A "sorority" of females	CONDITIONAL	At least 10 gal with cover. The group-housing research is about fish reared together from hatching
Another male betta	NEVER	They fight, often fatally
Fin-nippers, such as tiger barbs	NEVER	They shred the long fins that make a betta what it is
Long-finned fish, and goldfish	NEVER	A betta may spar with long fins. Goldfish need cooler water (65 to 75°F, 18.5 to 24°C) and are voracious fin-nippers

The four checks

A pairing has to pass all four: the temperature ranges overlap, the water chemistry overlaps, neither adult fits in the other's mouth, and neither is a known nipper or bully toward the other's body shape. The pairings that go wrong pass three. The temperature matched, the water matched, the sizes matched, and the barbs took the betta's tail apart in a week.

Adding a tankmate

Quarantine the newcomer first. Rearrange the decor while the bag floats, which breaks up the established territorial markers, and feed the betta at the moment the new fish is released. Then watch the betta's reaction: it is the thing to read.

WHEN IT IS FAILING, SEPARATE THEM

Flared gill covers and spread fins at a newcomer are a display. A display that turns into chasing and torn fins, on either fish, means the pairing has failed and one of them needs another home. Do not wait to see.

Males, females, bubble nests & fry

How breeders do it, and how little of it a pet owner needs.

Telling the sexes apart

Males and females only become distinguishable at about 3 to 4 months old. Breeders rear young bettas together as a group until then, and the males must be separated into their own containers at that point, because a male spreads his fins and flares his gill covers almost at once when he meets another male. Females are reared in communal tanks. Males are the ones with the flowing, showy fins and vivid color, which is why shops price them higher.

Bubble nests

A bubble nest is a raft of saliva-coated bubbles a male builds at the surface. It lets eggs develop in water that is warm, low in pH and low in dissolved oxygen, relatively free of predators. A male builds one with no female in the tank, and he keeps adding bubbles when the nest needs them, so a nest on its own is not a sign of breeding and is not a reason to worry. A healthy, content male will often maintain one indefinitely.

How breeding works

Breeding is a managed routine, not something that happens by accident in a pet tank. It is also a separate project from keeping a betta: it needs its own containers and a plan for every fry.

Setting up	A single male and female go into a container with floating vegetation or a small piece of foam, where the male builds the nest
Spawning	Typically within 1 to 3 days. The male wraps his body around the female and fertilizes the eggs as she releases them, a few at a time, about 300 eggs in a spawning
The nest	He catches the eggs in his mouth as they fall and places them in the nest, then tends them and adds bubbles when needed
The female	She is removed when spawning is finished
The fry	Removed to their own tank once they have absorbed their yolk sacs, about 3 days after hatching. First foods are infusoria (very small aquatic organisms) and newly hatched brine shrimp

The fry then follow the feeding stages on the diet page (page 15): 4 to 5 small meals a day once free-swimming, 3 to 4 a day by weeks 3 to 4, 2 to 3 a day as juveniles. Juveniles move onto the adult routine once they are separated into their own space.

What the aggression research means for you

REARING SETS TEMPERAMENT

The timing of isolation from an enriched environment shapes later aggression and the age of sexual maturity. A betta's temperament was set before you bought it, and none of that research makes two adult males safe together.

Red flags & testing the water first

Almost every common betta illness traces back to water quality or water temperature, so the first move is nearly always the same.

TEST THE WATER BEFORE YOU TREAT ANYTHING

Ammonia, nitrite, nitrate and temperature come first, before a single dose of medication goes in. Ammonia and nitrite are invisible, and a very large share of sudden "sick fish" presentations trace back to ammonia, nitrite or low oxygen rather than disease. Medicating before you have ruled that out can add stress to a fish already struggling with its environment. Correcting the water is not doing nothing while you wait: it is the treatment.

Act today

WHAT YOU SEE	WHAT IT POINTS AT	PAGE
Any ammonia or nitrite reading above 0 ppm	A water emergency. Change a third or more of the water now	10
Scales standing out like a pinecone, or a fish stuck floating, sinking or listing	Dropsy, a sign of organ failure with a poor prognosis, or swim bladder disorder	33
Grains of white salt, or a fine gold or rust dust, over the body	Ich, or velvet, which moves fast	31
Pale or grayish saddle-shaped patches, ragged white fin edges, a cottony fuzz	Columnaris, highly contagious and fast. Act the same day	32
Fins visibly shorter or more ragged than last week	Fin rot, if the damage is advancing	30
Gasping at the surface constantly, gills working hard	Ammonia or nitrite poisoning, or very low oxygen. Test immediately	34
Open sores or ulcers, or a curved or S-shaped spine	A vet's job. Rare fish tuberculosis can pass to people, so use gloves with any cut on your hands	33
Refusing food 5 to 7 days, or any refusal alongside another symptom	Investigate rather than wait	18

Finding a vet & when it is a vet's job

Most betta problems are solved at the test kit. The ones that are not need somebody who has seen a fish before.

When it is a vet's job

Call in an aquatic veterinarian or fish-health specialist when illness is spreading across more than one fish; when you see an open ulcer, significant bloating or a badly protruding eye; when fin erosion or a white-spot outbreak is advancing day over day despite a clean water test; when a fish is listing on its side or resting on the bottom unresponsive; or whenever the honest answer to "what caused this" is "I don't know, and the water tested clean." For many parasites, only a microscope on a sample confirms the diagnosis, which no owner can do from outside the tank.

Finding a vet who treats fish

Most small-animal clinics do not see fish, and a general search will not say which do. The American Association of Fish Veterinarians keeps a find-a-vet map that covers North America only. If it shows no one within reach, there are three next moves: ask a local vet, ideally one who already sees your other pets, whether they would take the case with guidance from a fish vet (the association will connect the two); contact the association directly, and it will email its members to look for someone near you; or ask an aquarium shop or a local aquarium society, which usually know who in the area treats fish. Do this now, while nothing is wrong, and write the number on the emergency card (page 37).

WHAT TO HAVE READY WHEN YOU CALL

Ammonia, nitrite, nitrate and pH, and the kit used. The temperature. Tank volume and how long it has run. The last water change. What the fish is doing, what it eats, anything added recently, and any medication already used, at what dose.

Fin rot, regrowth & clamped fins

A ragged fin is a symptom with several causes, and the fin tells you more over a week than in a single look.

Fin rot

What it is	A bacterial infection that shows as frayed, ragged or discolored fins
Treatment	Mild cases: clean water and aquarium salt, dosed for scaled fish only. Add antibiotics if it is more advanced or not improving, which is a vet's call (page 35)

The complication is that a ragged fin has several possible causes. Infection is only one of them: poor water quality, abrasion against unsuitable surfaces, disease and rough handling all produce the same frayed margin. Fin erosion is also a secondary consequence of bacterial and fungal disease, so damage that began as a torn fin can end up infected later. In a single betta tank the list is short: sharp decor, a filter intake, and water left too long between changes.

Direction of travel

Because the appearance overlaps, the reliable test is direction of travel. Photograph the fish from the side in good light, then compare the same view a week later. Damage that is holding still, or slowly filling back in, is wear. Damage that has moved further toward the body is active and needs treating. One picture cannot tell those apart, which is why fin rot gets diagnosed so often in fish that do not have it.

New growth looks like damage

Fins regrow, and the new tissue arrives without its color. In zebrafish, where regeneration has been measured most precisely, new tissue appears beyond the injury within a day or two and stays whitish for several days. Color returns from the base of the fin outward, and the fin's size and shape are restored after roughly three weeks. The clear or pale band sits at the outer edge the whole time, exactly where an owner has been told to watch for rot. A thin translucent margin on an otherwise clean fin is usually the fish repairing itself, and treating it as an infection works against the thing that closes the gap.

Age changes the timeline

Regrowth slows with age. In turquoise killifish, a species kept as a laboratory model for aging, young fish regrew 98 percent of an amputated tail fin within four weeks, middle-aged fish 78 percent, old fish 57 percent and the oldest 46 percent. A betta past its second birthday is genuinely old. Split and thinning fins on a fish that eats well, swims normally and holds its color are consistent with age, and clean water still helps, but it stops being a cure.

CLAMPED FINS ARE A POSTURE, NOT DAMAGE

Clamped fins are held shut and pressed against the body instead of spread and moving. They travel with signs that affect the whole fish, such as lethargy and darkened color. A fish with torn fins held fully open is showing wear on the tissue. A fish with undamaged fins held clamped is showing distress, and the water is the first place to look.

Ich & velvet

Two parasites, one of which kills in days. Know the difference by the look of the dust.

Ich (white spot disease)

Cause	The parasite <i>Ichthyophthirius multifiliis</i> , usually brought in on a new fish and set off by a chill or a stress event
Signs	Small white grains, evenly scattered like salt, across the body and fins, with scratching against decor. Clamped fins and lethargy often go with it
Why timing matters	The white spot is the one stage you cannot kill: the fish's own skin armors it, and the cyst on the gravel is armored by its wall. Only the free-swimming stage can be killed, so every treatment keeps the water lethal long enough for that stage to die before it finds a fish
Warmer is faster	At 75 to 79°F (24 to 26°C) the whole cycle completes in about 3 to 6 days. Colder stretches it to weeks

Treating ich in a betta

1. **Heat.** A betta can take the heat method. Raise the tank slowly, a couple of degrees a day, to 86°F (30°C) with an air stone running, since warm water holds less oxygen, and hold it for two weeks. That is a treatment ceiling for the course only, above the 76 to 82°F a betta lives at.
2. **Or salt.** About a tablespoon of aquarium salt per 5 gal, added gradually so the fish can acclimate. This is for scaled fish, and it is not for shrimp or snails.
3. **Or medication,** for a heavy outbreak: a malachite green or formalin product, used to its label or a vet's direction, in the hospital tank, with shrimp and snails out first.
4. **Keep going.** Continue until every spot has been gone for days and no fish has died from it. Stopping when the spots vanish leaves the cysts in the gravel to hatch a fresh wave.
5. **During treatment:** vacuum the gravel with each water change, pull the carbon, aerate, and count the fish daily.
6. **After:** quarantine for 30 days (page 25).

Velvet (gold dust disease)

Cause	<i>Piscinoodinium</i> , the freshwater velvet parasite. <i>Oodinium</i> is its marine counterpart, and the two names get swapped constantly
Signs	A fine gold or rust-colored dusty sheen over the body
Why it is urgent	It moves fast and can kill quickly: freshwater <i>Piscinoodinium</i> infections can cause high mortality once established
Response	Darken the tank and use an antiparasitic or copper treatment, started early, in the hospital tank (page 35 has the rules)

DO NOT GUESS BETWEEN ICH AND COLUMNARIS

The temperature moves point in opposite directions: ich is treated by warming the tank, while columnaris calls for cooling toward the low end of the betta's range. Getting it backward accelerates the infection.

Columnaris

A bacterial infection that moves in hours to days, and the one illness where the heater is the wrong reflex.

Cause	<i>Flavobacterium columnare</i> , a bacterium that is highly contagious between fish, and between tanks on shared equipment
Signs	Pale or grayish saddle-shaped patches over the back, ragged white edges on the fins and mouth, and sometimes a cottony fuzz that gets it mistaken for a fungus. It is not a fungus, and it moves far faster than fin rot
Virulence	Strains range from low to high. Mortality can be acute, and by the time it is visible the outbreak may already be hard to control, so act the same day rather than watch for a week

The temperature rule runs the other way

If you have already treated ich, the temperature move is the opposite one. Ich is managed by warming the tank, because the parasite runs its life cycle faster in warm water, so warming brings it sooner to the free-swimming stage that medication can reach. Columnaris runs the other way. In a review of the disease in *Veterinary Research*, Declercq and colleagues report mortality rising steeply with water temperature: none at 49°F (9.4°C), 4 to 20 percent at 54°F (12.2°C), and up to 100 percent in some species at 69°F (20.5°C), with the bacterium's grip on gill tissue enhanced as the water warms.

NEVER

Reach for the heater because it worked on the white spots last time. It is the worst thing you can do to a betta with columnaris. Cooling toward the low end of a betta's safe range, and not below about 76°F (24.5°C), is the move instead, and it only buys time for the antibacterial treatment: it is not a cure.

What to do

1. Test the water first, as for any illness.
2. Move toward the low end of the range slowly, and never below about 76°F.
3. Isolate the fish in the hospital tank.
4. Treat with an antibacterial medication, or veterinary antibiotics, dosed to the label or a vet's direction.
5. Disinfect nets, siphons and buckets before they touch another tank.

NARROW BAND, NARROW ROOM

A betta is a tropical fish with a narrow comfortable band, so there is less room to move the temperature in either direction than there is for a goldfish. That is why getting the direction right matters more here than it does in a coldwater tank.

Swim bladder, dropsy & fish tuberculosis

The internal problems, and the body signs that mean something is wrong inside.

Swim bladder disorder

Cause	Usually overfeeding, constipation or infection
Signs	Buoyancy problems: floating, sinking, or swimming at odd angles
Response	Fast the fish for 2 to 3 days, which often resolves mild cases. Afterward, offer a single cooked, de-shelled pea to help things move along. Buoyancy problems that do not clear after the fast need a vet
After	Cut the portion for good (page 19). Overfeeding is the usual cause, so it comes back otherwise

Dropsy

Raised, pinecone-like scales are the telltale sign. Dropsy is not really a single disease: it is a visible symptom of organ failure, from fluid building up inside the body. By the time the sign shows, the underlying problem is usually advanced and the prognosis is poor, so it is a same-day call to an aquatic vet rather than something to treat at home.

Mycobacteriosis (fish tuberculosis)

Rare, but worth knowing because it is zoonotic: it can pass to people through contact with infected water, especially through broken skin. Use gloves if you have any open cut and are doing maintenance on a tank you suspect is affected.

Other body signs

SIGN	WHAT IT CAN MEAN
Color loss, a washed-out look	A general illness indicator that often shows before anything more specific
Grayish-white, cotton-like growths	A fungal infection, most often on damaged skin, a healing wound or already-stressed tissue
Ulcers or red, inflamed patches	A bacterial process
Swollen belly	Overfeeding and digestive backup, dropsy-like fluid retention, or internal worms that give a pot-bellied look
Eyes visibly protruding or enlarged (popeye)	Associated with bacterial infection. Not specific enough on its own to name the cause
Scales standing out from the body, or lost scales	Scales standing out is the pinecone look of dropsy. Lost scales are one of the signs to track

SEVERAL SIGNS TOGETHER

By the time several of these appear together, a fish has usually been unwell for a while. Test the water, isolate the fish in the hospital tank, and call a vet if the signs spread or advance.

Reading waste, breathing & oxygen

Three signals a keeper can see without a microscope, and what each can and cannot tell you.

Waste: the earliest signal

Normal fish waste is a solid, somewhat compact strand that roughly matches the color of what the fish has been eating, and it does not trail behind for more than a few seconds before it drops or breaks off. The signal to watch is a shift to pale, clear or white waste that trails behind the fish in a long, stringy thread.

It is an early warning, not a diagnosis. Weight loss and a pot-bellied look are signs of internal worms, while weight loss and a refusal to eat are signs of an intestinal parasite. A fish that has simply stopped eating for a stretch, for any reason including a water problem, will often pass mostly mucus, which looks identical from outside. A single odd dropping in an eating, active fish is worth noting and rechecking, not panicking over. A pattern over several days, especially with weight loss or lost appetite, is the point to isolate the fish in a hospital tank and watch it closely.

Gasping and the oxygen question

Gasping at the surface is one of the most commonly misread signs. It is easy to assume a respiratory infection or a gill parasite. Far more often it is plain physics: dissolved oxygen above 5 milligrams per liter (mg/L, which for water is the same as ppm) supports optimum fish health, most species become distressed once it drops to roughly 2 to 4 mg/L, and deaths follow below about 2 mg/L, with surface gasping ("piping") as the visible warning. A betta has its labyrinth organ and takes air at the surface routinely. Constant, urgent gasping, with the gills working hard, is not that.

- A fish piping after a heat wave, a power outage that stopped the filter, or an overfed, overstocked tank is telling you about the water, not about a pathogen.
- A fish that hangs in the current from the filter outflow or an air stone is often chasing the most oxygen-rich water in the tank, so test the water.
- Nitrite converts hemoglobin into methemoglobin, the condition known as brown blood disease, and affected fish congregate in oxygen-rich water or gasp at the surface.
- Ammonia is toxic: the toxic form above roughly 0.05 mg/L is already enough to damage gill tissue, and acute ammonia poisoning can cause spinning and disorientation that looks like an exotic disease.

WHAT TO DO

Test ammonia, nitrite and nitrate and read the thermometer. Change water by the rules for the reading (page 12) and add surface agitation. Clear water does not mean safe water.

Behavior signs, in one list

Flashing, where the fish rubs against objects to relieve itching. Piping at the surface. Loss of appetite, loss of condition, dulled color and rapid breathing. Clamped fins. Abnormal buoyancy. Isolation or hiding in a fish that is normally active. Fish hide illness as a survival instinct, so a keeper who looks only once in the morning and once at night is working from incomplete information. Thirty seconds of watching, every day, is the habit that catches these early.

Medication rules & when a fish will not recover

The rules that apply whichever product you use, and the kindest option when treatment is past its use.

The universal rules

RULE	WHY
Treat in the hospital tank	Several medications do not tell the pathogen from the bacteria running the filter, and copper is extremely toxic to invertebrates and plants, which must come out before treatment
Pull the carbon first	Activated carbon adsorbs the medication out of the water before it can work. Take it out before dosing and keep it out until the course is finished
Turn up the aeration	Several treatments cut the oxygen. Vigorous aeration is essential with formalin and with the raised temperature of an ich treatment
Dose to the label or a vet	Every product and condition has its own concentration and course length. Treat any number you read elsewhere as a starting point to verify
Watch the tank, not just the fish	A hospital tank's filter is smaller and newer than the main tank's. Check ammonia and nitrite partway through a course

Aquarium salt: the limits

Aquarium salt is plain sodium chloride sold for aquarium use, not table salt. The starting dose for ich is about 1 tablespoon per 5 gal, added gradually so the fish can acclimate, and it is for scaled fish only. Catfish such as corydoras, loaches and shrimp do not tolerate it, which is a reason not to dose a tank with those tankmates in it. Other uses of salt are written in different units and are different jobs, so never convert one dose into another's.

- There is no fixed number of days. Once the infection has cleared, several water changes remove the salt, rather than a course that runs a set time regardless of how the fish is doing.
- Salt does not evaporate, so top off with plain, conditioned water, and remove salt only by a real water change.

When a fish will not recover

Some fish will not recover, and a fish vet is the one to judge whether a case is past treatment. Once it is, ending the fish's life humanely is part of keeping it. The first choice is a vet who treats fish: an overdose of anesthetic is the humane method, and a vet has a range of anesthetics and the training to use them. Where no vet can be reached, the method for aquarium fish is a clove oil overdose, since clove oil contains the sedative eugenol. Use about 0.4 milliliters (ml) of clove oil per liter of water, in a container of water taken from the fish's own tank. Mix the oil with a little warm water first and add it slowly over about five minutes. The fish loses consciousness, stops breathing and dies. Leave it in the solution for at least 30 minutes after the last gill movement. Only then, as a final check, can it go in the freezer.

METHODS TO RULE OUT

Flushing, which puts a live fish into chlorinated water. Freezing a live fish, boiling, cutting off the head without first stunning the fish, and leaving it out of water to suffocate. All cause the suffering the exercise is meant to end.

Setup checklist & targets

Essential setup checklist

- ☐ 5 gal (19 L) tank minimum, 10 gal (38 L) better, with a tight-fitting lid (page 6)
- ☐ Adjustable submersible heater, 15 to 25 W for 5 gal (page 7)
- ☐ A thermometer to read the tank with
- ☐ Sponge filter and air pump, flow kept gentle (page 8)
- ☐ Liquid reagent test kit for ammonia, nitrite, nitrate and pH, not strips (page 11)
- ☐ Water conditioner labeled for chlorine and chloramine (page 12)
- ☐ Soft substrate: sand or smooth gravel, and a gravel vacuum (page 13)
- ☐ Live or soft silk plants, hides, and a surface resting spot
- ☐ A light on a timer, with dark nights
- ☐ A net and a cup for transfers (page 20)
- ☐ Betta-specific pellets (page 15)
- ☐ A bare 10 to 20 gal hospital tank plan, with its own net (page 25)

Water targets

READING	TARGET	ACT IF
Temperature	78 to 80°F (25.5 to 26.5°C)	Outside 76 to 82°F (24.5 to 28°C)
Ammonia	0 ppm	Anything above 0
Nitrite	0 ppm	Anything above 0
Nitrate	Under 20 ppm	20 ppm or more before the weekly change
pH	6.5 to 7.5	Swinging, which is worse than sitting slightly off target
Water change	25 to 30% weekly, conditioned for chlorine and chloramine and matched to the tank's temperature, siphoned from the substrate	
Light	6 to 8 hours a day in a newly planted tank, 8 to 12 once the plants grow, dark at night	
Feeding	2 to 4 pellets, once or twice a day, one fasting day a week	

Before your betta comes home

- ☐ The tank is fully cycled: a full ammonia dose reads zero within 24 hours, with nitrite at zero and nitrate present
- ☐ Nitrate brought under 20 ppm with a water change
- ☐ The heater has run long enough to hold a steady 78 to 80°F (25.5 to 26.5°C)
- ☐ A vet who treats fish found, and the number written on the emergency card

Emergency & quick targets

ACT TODAY IF

Ammonia or nitrite above 0 ppm · scales standing out like a pinecone · a fine gold or rust dust over the body · pale or gray saddle-shaped patches or cottony fuzz, spreading · fins shorter or more ragged than last week · white grains across body and fins · stuck floating, sinking or listing · constant urgent gasping at the surface · open ulcers, or a badly protruding eye · 5 to 7 days of refused food, or any refusal with another symptom

READING	TARGET
Temperature	78 to 80°F (25.5 to 26.5°C). Act outside 76 to 82°F (24.5 to 28°C)
Ammonia and nitrite	0 ppm. At 0.1 ppm change at least a third of the water; 50% or more once a reading is clearly climbing
Nitrate	Under 20 ppm before the weekly change
pH	6.5 to 7.5, stable
Feeding	2 to 4 pellets, once or twice a day, one fasting day a week
Water change	25 to 30% weekly, conditioned and temperature-matched
Ich	Slowly to 86°F (30°C) with an air stone, held two weeks, or a tablespoon of aquarium salt per 5 gal
Columnaris	Cool toward the low end of the range, never below about 76°F (24.5°C), and isolate

Your numbers

Vet who treats fish _____

Tank volume and heater watts _____

Aquarium shop or society _____

Date the tank finished cycling _____

Fish's name and arrival date _____

Last water change _____

THE FIRST MOVE

Test the water and read the thermometer before anything else. A matched, conditioned water change is the right first response to most water emergencies, and correcting the water is the treatment, not a delay.

Budget: setup & shopping list

United States retail ranges, last checked October 2026 and rounded to the nearest \$5. Prices move; treat this as a snapshot, not a quote.

INITIAL SETUP	TYPICAL COST
5 gal or larger tank with a tight-fitting lid and a light, often sold as a kit; check the dimensions before you buy	\$30 to \$100
Adjustable submersible heater, 15 to 25 W for 5 gal; check the wattage against your tank's volume	\$15 to \$30
Sponge filter, the gentle standard; check its rating against your tank's volume	\$5 to \$15
Small air pump with airline and a check valve, to run the sponge filter	\$10 to \$15
Digital aquarium thermometer for the water	\$10 to \$15
Decor: a smooth hide, live or soft silk plants, and a resting spot near the surface	\$10 to \$50
Substrate: sand or smooth gravel; check the quantity against your tank's floor	\$10 to \$20
Water conditioner, one 16-ounce bottle	\$5 to \$15
Liquid reagent test kit, one-time and reusable	\$15 to \$40
Small gravel vacuum sized for the tank	about \$10
Plug-in timer for the light	\$5 to \$15
Soft fine-mesh net, with any clean cup for transfers	about \$5
First supply of betta-specific pellets	\$5 to \$15
First supply of freeze-dried bloodworms for variety	about \$15
Setup total, every row added up, before the fish	\$150 to \$360
The betta itself (up to about \$55 for a rare variety)	\$5 to \$30
All in, with a \$5 to \$30 fish	\$155 to \$390

Males cost more than females, mostly for their showier fins and color, not for any difference in care.

NOT ON THE LIST

A bare hospital tank for quarantine and treatment is a second tank, with its own sponge filter, heater and net, so price it like the first tank's rows (page 25). Vet care for a betta is rare: most health issues are managed by correcting water quality, though an aquatic vet exists for serious cases.

Budget: running costs

What a betta costs to keep each year once the setup on the previous page is in place, on the same rounding.

YEARLY ONGOING	TYPICAL COST
Food top-ups: pellets and freeze-dried treats	\$30 to \$50
Electricity for the heater, air pump and light	\$30 to \$100
Everything else: conditioner top-ups, replacement filter media, test reagent refills	\$60 to \$150
Yearly total, every row added up	\$120 to \$300
Monthly equivalent	\$10 to \$25

A year that needs a course of medication runs higher than the top of that range. Food is almost never the expensive part, since a betta eats a few pellets a day.

WHERE NOT TO CUT CORNERS

The heater and the filter. The cheap bowl-and-fish bundle is missing exactly those two, and spending the extra money upfront on a real tank with a heater is the difference between a fish that struggles and one that thrives. Save on decor first.

First 30 days

Day 1 is the day the fish arrives, which is weeks after the day the tank was set up.

Before day 1

- ☐ Tank cycled and confirmed: a full ammonia dose reads zero in 24 hours, with nitrite zero (page 9)
- ☐ Heater steady at 78 to 80°F (25.5 to 26.5°C) on your own thermometer
- ☐ Vet who treats fish found, and the number on the emergency card (page 37)

Day 1

- ☐ Float the sealed cup or bag 20 to 30 minutes, then net the fish across. Never pour the store water in (page 24)
- ☐ Let the fish settle: stress can suppress appetite for a few days
- ☐ Start the equipment dates (page 47)

Week 1

- ☐ Test ammonia and nitrite every 1 to 2 days: a new fish is a new load (page 11)
- ☐ Offer the measured portion, and expect a refused meal or two (page 18)
- ☐ Watch for flashing, clamped fins, white spots or a gold dust
- ☐ No tankmates, and no rearranging

Weeks 2 to 3

- ☐ Settle into the feeding routine, including the weekly fasting day (page 15)
- ☐ First full weekly change, 25 to 30%, conditioned and matched (page 12)
- ☐ Start the Owner Log (page 46) and take the first top-down look at body shape
- ☐ Move the feeding spot around the tank

Week 4

- ☐ Drop testing to weekly once ammonia and nitrite have read zero for a full week
- ☐ Test nitrate before the weekly change, and let the reading set the size of the change
- ☐ Add planting or cover, one change at a time (page 48)

Days 31 to 90

- ☐ Read the Owner Log for slow trends, not single readings
- ☐ Fill in the pet-sitter sheet before you need it (page 45)
- ☐ Put together the power-outage kit (page 43)

Symptom quick reference

SYMPTOM	LIKELY CAUSE	FIRST ACTION	PAGE
White grains like salt across body and fins	Ich	Slowly to 86°F (30°C) with an air stone, or aquarium salt	31
Fine gold or rust dust	Velvet	Darken the tank, antiparasitic or copper, act today	31
Pale or gray saddle patch, ragged white edges, fuzz	Columnaris	Cool toward 76°F (24°C), isolate, antibacterial, same day	32
Fins frayed, ragged or shorter than last week	Fin rot, if advancing	Fix the water, photograph weekly	30
Thin pale margin on an otherwise clean fin	Regrowth	Keep the water clean, do not medicate	30
Scales standing out like a pinecone	Dropsy, organ failure	Isolate, pristine water, call an aquatic vet	33
Floating, sinking or listing	Swim bladder disorder	Fast 2 to 3 days, then one cooked, de-shelled pea	33
Bloated belly, stringy or absent feces	Overfeeding or constipation	Fast, check the temperature, pre-soak pellets	17
Pale or white stringy waste	A nonspecific sign	Watch the fish for a day or two, then isolate if it repeats	34
Constant urgent gasping at the surface	Ammonia, nitrite or low oxygen	Test now, change water, add agitation	34
Flashing against decor	An irritant or a parasite	Test, then look closely for white grains or gold dust	21
Clamped fins, faded color, lethargy	Stress, usually water or temperature	Test the water and the temperature first	28
Refusing food, otherwise normal	New tank, cold water or picky	Work the six-reason list, then wait	18
Refusing food 5 to 7 days with a second sign	Several possible causes	Investigate now, bring the Owner Log	46
Hiding constantly in a bare tank	Restricted space and no cover	Add volume, planting and cover	22
Torn fin from decor, with a clean edge	Abrasion, not infection	Remove the sharp decor, keep the water clean	13
Chasing and torn fins after a tankmate went in	A failed pairing	Separate them at once	26
Ammonia or nitrite above 0 in a settled tank	A broken cycle	Work back through the stall list, change water	10

Daily, weekly & seasonal routine

Daily

- ☐ Read the thermometer before anything else
- ☐ Feed the measured portion, moving the spot around, and remove anything uneaten after 1 to 2 minutes
- ☐ Thirty seconds of watching: fins, color, breathing and how much of the tank it uses
- ☐ Check the filter is running and the surface is moving
- ☐ Check the light timer switched on and off as it should

Weekly

- ☐ Test ammonia, nitrite and nitrate on the same day each week, nitrate before the change, and log it (page 46)
- ☐ Change 25 to 30%, conditioned and matched, vacuuming a different section of substrate (page 12)
- ☐ One fasting day
- ☐ A top-down look at body shape (page 19)

Monthly

- ☐ Rinse the filter sponge in old tank water only if flow has dropped, never under the tap (page 14)
- ☐ Check the heater holds its setting against the thermometer
- ☐ Read the log for slow trends rather than single readings
- ☐ Rotate one piece of enrichment (page 48)

Seasonal

- ☐ Winter: the room cools, so check the heater is keeping up and the tank is still 76 to 82°F (24.5 to 28°C)
- ☐ Summer: in a hot room warm water holds less oxygen, so add surface agitation if the tank runs hot, and keep it out of direct sun (page 7)

Every treatment or illness

- ☐ Keep testing ammonia and nitrite after the course ends (page 35)

Power outages

An aquarium looks like it depends on light. It depends on motion: the filter and air pump keep the surface moving, and that is what brings oxygen in. Aerate before you do anything else.

What kills fish in an outage

Oxygen depletion is the fast risk. Once the filter stops, the surface stops moving, gas exchange slows, and dissolved oxygen starts falling within a couple of hours, sooner in a warm tank. Temperature drift is the slower risk: most home tanks are insulated by their own volume and the room, so a few hours without power rarely swings the water enough to be dangerous. Waste is the risk you create by getting either of the first two wrong.

Do

- ☐ Aerate right away. A battery-powered air pump with an air stone is the cleanest fix
- ☐ No pump: scoop water out with a cup and pour it back from a few inches up for a minute or two, roughly every 10 to 15 minutes in a small tank
- ☐ Insulate with towels or blankets and keep the lid on, rather than adding any new heat source near water and electronics
- ☐ Unplug the filter at the wall, so you control when it comes back
- ☐ Check on the tank regularly. A fish gasping at the surface means aerate more often

Don't

- ☐ Feed. Uneaten food consumes oxygen and rots in still water
- ☐ Restart a filter that has been off for hours without checking it
- ☐ Add a space heater, heat lamp or open flame near the tank
- ☐ Panic-swap large amounts of water. A big change adds stress, and a second problem if the new water is not matched

When the power comes back

A filter that has sat dead for hours is a small sealed pocket where oxygen ran out and waste kept decomposing. Switch it on without checking and all of that is pushed into the tank at once: an ammonia spike and a slug of oxygen-poor water. Open the filter and rinse the media in a bucket of the tank's own water, never tap water. If the outage ran a full day or more, test ammonia and nitrite once things are running again and keep testing for a few days, since a weakened colony can take time to catch up.

BLACKOUT KIT

A battery air pump with a spare air stone, a cup or jug for the pour-back method, two thick towels, a thermometer and the numbers from the emergency card (page 37).

COAST, BUT AERATE

A well-established tank can generally coast through several hours without heat or filtration as long as oxygen is being managed, and the heater's 76 to 82°F (24.5 to 28°C) range is the number to watch. Aeration is the whole job.

Travel, transport & moving

A betta travels differently from most fish, because it breathes the air in the bag.

Leaving town

TRIP	WHAT TO DO
A weekend	Low risk for a healthy adult with an automatic feeder. A healthy adult can survive up to about two weeks without food, but that is an emergency ceiling, not a target
Longer	A sitter, with the meals pre-portioned and the sitter sheet filled in. Refusal of food for 5 to 7 days is the point to investigate

Bagging a betta

A bag beats a rigid bucket for most short trips, such as a local drive or a vet visit: it holds the fish in the dark, which is calming, and it flexes instead of sloshing on turns. A bucket or tub with a tight lid earns its place on longer trips, since it is harder to puncture and easier to aerate with a battery pump.

- Leave headspace. For a betta the air gap matters even more than for most fish, since it takes air directly. Air should be the larger share. Commercial betta bags run 90 percent air with only enough water to keep the fish submerged.
- For a short drive at home, fill the bag no more than a third with water and leave the rest as air.
- For a longer trip, pure oxygen is standard practice: it saturates the water with far more dissolved oxygen than air can.
- Keep the bag out of direct sun and never in a hot car. After several hours, start thinking about an insulated box around the bag.

Moving the tank

A full move is the outage and the transport problem at once, plus one more piece: the filter's colony has to survive being out of the tank. That colony lives mostly on the media, so the media can travel separately, as long as it stays wet.

- Pack the sponge in a sealed bag or tub of the tank's own water, never tap water, and never dry. For a longer drive, drop a battery air stone into the same container.
- Set the filter up as soon as the new tank is filled, using that reserved water, so there is no gap where the colony sits idle. Handled this way, most moves do not need a fresh cycle.
- Move the fish last, after the new tank's water is conditioned and close to the old tank's temperature, and get it out of the bag as soon as you reasonably can.

THE FIRST WEEK AFTER A MOVE

Test ammonia and nitrite for a few days: a weakened colony can need time to catch up, and a new room is a new ambient temperature, so read the thermometer too. Keep the tank out of direct sun and away from drafts.

FILL IN AND LEAVE BESIDE THE TANK

Pet-sitter sheet

Everything a sitter needs on one page. Walk them through it once in person before the first solo visit.

Fish's name _____

Vet who treats fish _____

Tank volume and where the food is _____

Backup contact _____

My phone _____

Days to feed on _____

Each visit

- ☐ Read the thermometer. It should say 78 to 80°F (25.5 to 26.5°C). Write it down
- ☐ Check the filter is running and the water surface is moving
- ☐ Find the fish. It should be swimming, with its fins open. Coming up to the surface for air is normal
- ☐ Feed only on the marked days, and only the pre-portioned amount. This is a very small amount of food, and that is correct
- ☐ If anything is floating uneaten after a couple of minutes, scoop it out with the net

Please don't

- Feed extra, or feed on an unmarked day. Overfeeding is the commonest mistake with this fish, and it will always look hungry.
- Add tap water, change water or top the tank up unless I have written instructions. Untreated tap water is harmful here.
- Change any dial or setting on the heater, the pump or the light timer.
- Tap the glass, hold up a mirror, or put anything into the tank.
- Take the lid off and leave it off. Bettas jump.

CALL ME IF

The thermometer reads outside 76 to 82°F (24.5 to 28°C) · the filter has stopped · the power has gone out (page 43) · the fish is stuck at the surface, on the bottom or on its side · its fins are clamped tight or ragged · it has white spots, a gold dusty sheen or white fuzzy patches · its belly is swollen or its scales stick out · the water has turned cloudy or smells bad · it has not eaten on a feeding day for several days

SECTION 08 · OWNER TOOLS

Owner log

One row a week, filled in on test day. Photocopy this page, or track digitally. Consistency matters more than the format.

[illegible]

WHY BOTHER LOGGING

You cannot weigh a betta at home, so your test results and a written description of the fish, with the top-down body shape, are the only trend data you will ever have. A nitrate reading creeping up week on week tells you the water change is too small before the fish does. Sixteen rows is four months, long enough to see a real trend rather than a bad day.

Equipment, water change & vet log

Equipment log

ITEM	INSTALLED	CHECK OR REPLACE BY	DONE
Heater (check it holds its setting against the thermometer)			
Air pump and airline			
Filter sponge (swap a portion at a time, never all at once)			
Activated carbon, if used			
Test kit reagents			

Twelve-month planner

Tick the box in the month you did it. Start with whichever month you are in.

TASK	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Check heater against the thermometer	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Review the log for trends	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Rotate one enrichment item	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Quarantine log (any new fish, 30 days minimum)

DATE IN	DATE OUT	WHAT IT WAS, AND ANYTHING NOTICED

Treatment & vet log

DATE	SYMPTOM OR DIAGNOSIS	TREATMENT, DOSE AND DURATION	OUTCOME

Enrichment checklist & log

Setup

- ☐ Dense planting, live where possible, soft silk where not
- ☐ A surface resting spot: a leaf hammock or a broad-leaved plant near the top
- ☐ Driftwood or decor placed to break the tank into zones and cut sightlines
- ☐ At least one hide, with nothing sharp to tear a fin

Monthly review

- ☐ Move or add one item, without blocking the way to the surface
- ☐ Vary the food and move the feeding spot, and try one new safe food from the food chart
- ☐ Check whether the fish uses more of the tank than last month, not less

Enrichment log

DATE	WHAT CHANGED	HOW IT WENT, AND WHETHER TO REPEAT

WHAT NOT TO DO

Do not use a mirror as a daily routine: it provokes a threat display and a measurable stress response. Do not use stiff plastic plants. Do not add so much decor that the fish cannot reach the surface, because air access outranks everything else on this page. And do not change three things at once, or you will never know which one the fish responded to.

REFERENCE

Glossary, A to F

Acclimate	To bring a fish gradually to a new tank's temperature and water chemistry, by floating the sealed cup and, where the waters differ a lot, adding tank water to the cup a little at a time. Page 24.
Air stone	A porous stone on an air line that releases fine bubbles. It adds oxygen and moves the water surface, which is how a tank takes in air. Pages 31, 35.
Ammonia (NH3)	The first and most toxic product of fish waste and decaying food. NH3 is its chemical formula: one nitrogen and three hydrogens. The target is 0 ppm, always. Pages 9, 11.
Aquarium salt	Plain sodium chloride sold for aquarium use, not table salt. A treatment dosed into the water for scaled fish, never a food. Pages 31, 35.
Brine shrimp nauplii	Newly hatched baby brine shrimp, a first food for fry. Page 27.
Bubble nest	A raft of saliva-coated air bubbles a male builds at the surface to hold eggs. Males build them with no female present. Pages 21, 27.
°C	Degrees Celsius, the metric temperature scale, on which water freezes at 0 and boils at 100. Page 7.
Chloramine	Chlorine bonded to ammonia, used by many water utilities. It does not gas off the way chlorine does, so a conditioner has to be labeled for both. Page 12.
Clamped fins	Fins held tight against the body instead of spread. A posture that signals stress or illness, not damage to the fin itself. Pages 21, 30.
cm	Centimeter, one hundredth of a meter, about 0.4 in.
Columnaris	A fast, highly contagious bacterial infection of fish, from <i>Flavobacterium columnare</i> , that makes pale or gray saddle-shaped patches. It is not a fungus. Page 32.
Conditioner	A water treatment, also called a dechlorinator, that neutralizes the chlorine and chloramine in tap water. Page 12.
Cortisol	A stress hormone. Researchers measure it after a mirror display to see how stressed a betta is. Page 21.
Cycled	A tank whose bacteria colonies keep pace with the waste: a full ammonia dose reads zero within 24 hours, nitrite reads zero, and nitrate is present. The single prerequisite for adding a fish. Pages 9, 10.
Dissolved oxygen	The oxygen mixed into the water, which gills take up. It falls as water warms and when the surface stops moving. Pages 7, 34.
Dropsy	Not a disease but a sign: fluid building up inside the body from organ failure, with scales standing out like a pinecone. Page 33.
Eugenol	The sedative in clove oil, which is why clove oil is the humane method where no vet can be reached. Page 35.
°F	Degrees Fahrenheit, the temperature scale used in the United States, on which water freezes at 32 and boils at 212. Page 7.
Fasting day	One day a week with no food, to support digestion and reduce bloating and constipation. Page 15.
Fin rot	A bacterial infection that frays and discolors the fins, often secondary to poor water, a chill or a physical fin injury. Page 30.
Fish-in cycle	Cycling a tank with the fish already living in it, using daily testing and large water changes to keep ammonia and nitrite low. A rescue, not a plan. Page 10.
Fishless cycle	Cycling an empty tank by dosing pure ammonia to feed the bacteria before any fish arrives. The safer path. Page 9.
Flashing	A fish rubbing or scratching against decor or the substrate, usually a sign of an irritant or a parasite. Pages 21, 34.

REFERENCE

Glossary, G to M

gal	Gallon, the United States unit of liquid volume, about 3.8 L. Page 6.
GH	General hardness: the amount of dissolved calcium and magnesium in the water, a mineral supply that does not buffer pH. Page 11.
Gravel vacuum	A siphon with a wide tube that pulls waste out of the substrate during a water change. Page 12.
Hospital tank	A bare tank with a seeded sponge filter, an air pump and a heater, used to quarantine new fish and to treat sick ones without dosing the main tank. Pages 25, 35.
Ich	White spot disease, caused by the parasite <i>Ichthyophthirius multifiliis</i> , which shows as grains of white salt across the body and fins. Page 31.
in	Inch, 2.54 cm.
Infusoria	A mixture of very small aquatic organisms, the first food of newly hatched fry. Pages 15, 27.
IUCN	International Union for Conservation of Nature, which keeps the global Red List of threatened species. The betta is listed Vulnerable in the wild. Page 4.
KH	Carbonate hardness: the water's buffering capacity, its ability to neutralize acid and hold pH steady. The bacteria that clear ammonia use it up. Pages 10, 11.
L	Liter, the metric unit of liquid volume, about 0.26 gal.
Labyrinth organ	A structure above the gills that lets a betta take oxygen from air at the surface. It is why clear access to the surface is a physiological need. Pages 5, 6.
Malachite green, formalin	Two standard medications for ich in fish that cannot take heat or salt, or in a heavy outbreak. Used to the label or a vet's direction, in a hospital tank. Page 31.
Methemoglobin	The form of the blood's oxygen carrier that cannot carry oxygen. Nitrite makes it, which is why nitrite poisoning is called brown blood disease. Page 34.
mg/L	Milligrams per liter, a weight of a substance in a volume of water, which for water is the same as ppm. For dissolved oxygen, 5 mg/L supports optimum fish health. Page 34.
ml	Milliliter, one thousandth of a liter. Page 35.
Mycobacteriosis	Fish tuberculosis: a rare bacterial infection that is zoonotic. Pages 14, 33.

REFERENCE

Glossary, N to Z

Nitrate (NO3)	The end product of a healthy cycle. Far less harmful than ammonia or nitrite, and removed by water changes. Keep it under 20 ppm. Pages 9, 11, 12.
Nitrite (NO2)	The middle step of the nitrogen cycle. It blocks oxygen uptake in the blood. The target is 0 ppm. Pages 9, 11.
Nitrogen cycle	The chain in which bacteria convert ammonia to nitrite and then to nitrate. Cycling a tank means growing both bacteria groups. Page 9.
Obligate carnivore	An animal that needs meat to live. A betta in the wild eats insects, larvae and small crustaceans. Page 15.
pH	The standard scale of how acidic or basic water is, from 0 to 14, with 7 neutral. For a betta, 6.5 to 7.5 and steady. Pages 10, 11.
Piping	A fish swimming at the surface to gulp air, a warning sign of low oxygen or poisoned water. Page 34.
ppm	Parts per million: the unit a test kit prints, one part of a substance in a million parts of water. Pages 9, 11.
Quarantine	Keeping a new or sick fish in a separate tank for a set time, at least 30 days, before it joins others. Page 25.
Seeded media	Filter sponge or gravel taken from an established, healthy tank, carrying a living bacteria colony. The most reliable way to shorten a cycle. Pages 9, 25.
Slime coat	The mucus layer over a fish's body, a barrier against infection and parasites. Bare hands strip it. Page 20.
Sorority	A group of female bettas kept together, which needs at least 10 gal and cover. Page 26.
Sponge filter	An air-driven filter whose sponge is the surface the bacteria live on. The standard for bettas because it filters with almost no current. Page 8.
Surface agitation	Movement of the water surface, from a filter, air stone or hand pouring, which is how oxygen enters the tank. Pages 7, 34.
Swim bladder	The gas-filled organ that controls buoyancy. A swollen gut pressing on it makes a fish float, sink or list. Page 33.
Tannins	Compounds released by driftwood that tint water faintly amber and soften the light. Cosmetic, and closer to the blackwater the species comes from. Pages 13, 22.
VCA	Veterinary Centers of America, a United States veterinary hospital group, named only on the Sources page.
Velvet	Gold dust disease, caused by the freshwater parasite <i>Piscinoodinium</i> , which dusts the body with a fine gold or rust sheen. Fast and dangerous. Page 31.
W	Watt, the unit of power. A heater of 15 to 25 W suits a 5 gal tank. Page 7.
Zoonotic	Able to pass from animals to people. Fish tuberculosis can, through broken skin. Pages 14, 33.

REFERENCE

Sources

Where the numbers in this package come from, so you can check them yourself.

Veterinary and extension references

- Merck Veterinary Manual: management of aquarium fish, routine health care of fish, overview of aquarium fish, providing a home for fish, disorders and diseases of fish, parasitic and bacterial diseases of fish, and the aquatic systems pages on life support and environmental disease. The 30 day quarantine minimum, the bare hospital tank, the carbon and aeration rules, the float time and bag-water rule, and the sign lists.
- University of Florida Institute of Food and Agricultural Sciences Extension: the ich (white spot) fact sheet; ammonia in aquatic systems; dissolved oxygen for fish production; on-farm transport of ornamental fish; and Evolution, Culture, and Care for *Betta splendens* (Tuckett, Ritch, Lawson and Hill). The ich life cycle, the oxygen thresholds, the slow-acclimation temperature gaps, the bubble nest and breeding routine.
- North Carolina State University College of Veterinary Medicine: dealing with aquariums and ponds during power outages. Southern Regional Aquaculture Center: how to start a biofilter, and management of recirculating systems. Aeration ahead of temperature, and filter media care.
- Jessie Sanders, a veterinarian board-certified in fish practice, at Fish Vet: how to fix a fat betta, old tank syndrome, and dechlorinating fish systems. The top-down body condition check and the pellet cut.
- PetMD: the betta fish care sheet, and nitrate poisoning in fish. Veterinary Centers of America (VCA) Animal Hospitals: fish white spot disease. The aquarium salt dose for ich.
- American Association of Fish Veterinarians: the find-a-vet directory. Royal Society for the Prevention of Cruelty to Animals Australia: the most humane way to euthanize aquarium fish.

Research

- Betta enrichment: *Life in a fishbowl* (Animal Welfare); *Life beyond a jar*; *Effects of aquatic plants on water quality, microbial community and fish behaviors in newly established Betta aquaria*; and the BioMed Central Zoology (2021) study on the timing of isolation from an enriched environment.
- Kajimura and colleagues (2023), BioMed Central Zoology: acute toxicity of ammonia and nitrite to Siamese fighting fish. Giannecchini, Massago and Fernandes (2012), Revista Brasileira de Zootecnia 41(4): photoperiod and reproduction.
- Declercq, Haesebrouck, Van den Broeck, Bossier and Decostere (2013), Veterinary Research 44:27: columnaris disease in fish. Pedrazzani and colleagues (2020), Frontiers in Veterinary Science 7:606388; Pfefferli and Jazwinska (2015), Regeneration 2(2); Wendler and colleagues (2015), Aging Cell 14(5): the causes of fin erosion, fin regrowth, and its decline with age.
- Microbial community succession of home aquarium biofilters (the International Society for Microbial Ecology's journal, 2025): how long real home filters take to cycle.

Husbandry references

- Aquarium Co-Op: the betta care guide, and its pages on water changes, ich, stocking and aquarium lighting. Practical Fishkeeping: the guide to cycling a tank and its questions on ammonia and fishless cycling.
- Bettafish.org: food and feeding. Dial-A-Vet: what to do if your betta is not eating (veterinarian reviewed). The Fish Site: transportation of fish in bags. Aqueon: how to move an aquarium.
- The Aquarium Expert and A-Z Animals: the itemized retail price pages behind the budget.

REFERENCE

Where the sources disagree

Where credible sources give different numbers, this package prints one and says why.

TOPIC	WHAT THE SOURCES SAY	WHAT THIS PACKAGE USES
Smallest tank	At least 2.5 gal, 5 to 10 gal ideal (a pet-health reference); males thrive in small containers when water quality is managed (a university extension paper on commercial culture)	5 gal minimum, 10 gal easier, because more water holds steadier
Meals a day	Once a day; once or twice a day; up to three smaller meals 6 to 8 hours apart (veterinary sources)	Once or twice a day, held consistent, plus one fasting day a week
Protein in the pellet	Roughly 30 percent or higher (a hobby feeding guide); about 35 percent with about 5 percent fat (a fish veterinarian)	30 percent or higher, about 35 percent and 5 percent fat as the target
How long a cycle takes	Four to six weeks (hobby and extension guides); about three weeks to eight in real home filters (a 2025 study)	4 to 6 weeks, and plan for up to 8
Fish-in action line	Act at 0.1 ppm with a change of at least a third; managed day to day at 0.1 to 0.25 ppm; 50 percent or more when a reading is clearly climbing	Act at 0.1 ppm, scaling the change up as the reading climbs
How much nitrate	Most fish tolerate about 20 ppm; 40 ppm is the ceiling below which most are considered safe	Under 20 ppm before the weekly change, and act past 40
A betta that will not eat	It can survive up to about two weeks; veterinary-reviewed guidance says investigate at 5 to 7 days	Investigate at 5 to 7 days, at once with a physical sign
Hospital tank size	A 10 gal home build; veterinary practices run 10 to 20 gal	10 to 20 gal, bare
Aquarium salt	About a tablespoon per 5 gal for ich; other doses for transport and for ongoing use, written in grams per liter	Only the ich dose, for scaled fish, and never converted from another row
Air in a betta's bag	Shipping bags run about 75 percent oxygen; commercial betta bags about 90 percent air	Air the larger share; a third water at most for a short drive
Female groups	A sorority is fine in at least 10 gal (a pet-health reference); the group-housing research is about fish reared together from hatching	Allowed at 10 gal with cover
Ghost shrimp with a betta	"Pose no threat" to the betta, but juveniles get eaten; safe once past about 1.5 in	10 gal or more, hiding places, shrimp past 1.5 in

REFERENCE

Version history & about

What changed in each edition, so a printed copy can be placed.

Version history

This is edition 3.1 (Oct 2026). The store page shows the current edition; re-download free if yours is older. Where this list and a page disagree, the page wins.

- **3.1, Oct 2026.** Now 54 pages. Costs: every item priced and rounded to the nearest \$5; setup \$150 to \$360, running costs \$10 to \$25 a month, the budget now over two pages. Other changes are wording only.
- **3.0, Oct 2026.** Rebuilt on the current outline, 53 pages. Safety: weekly water change 25 to 30%; float 20 to 30 minutes, slow mixing past a 10°F gap; ich by heat to 86°F (30°C) with an air stone for two weeks, or aquarium salt at a tablespoon per 5 gal; fishless cycle dosed to 3 ppm, done when a full dose reads zero in 24 hours, 4 to 6 weeks and up to 8; fish-in action line 0.1 ppm, a third of the water, 50% or more if climbing; hospital tank 10 to 20 gal, bare; tankmates need 10 to 20 gal, ghost shrimp past 1.5 in, sorority at 10 gal with cover; no heat source in an outage; a bag no more than a third water; pellet about 35% protein and 5% fat; aquarium salt for ich only, fin rot dosed to a label or a vet. Costs: setup \$105 to \$280 before the fish, \$110 to \$310 all in. Other changes are wording only.
- **2.2, Sep 2026.** Safety: quarantine 30 days minimum; salt ended with repeated water changes once the infection clears; aeration first in a blackout; fish-in action line 0.1 to 0.25 ppm. Other changes are wording only.
- **2.1, Sep 2026.** Spacing rules only. No words and no figures changed.
- **2.0, Sep 2026.** Safety: feeding window 1 to 2 minutes. Costs: running costs \$10 to \$25 a month, \$120 to \$300 a year.
- **1.0, Sep 2026.** First edition, 36 pages. Replaced by 2.0.

This package is a husbandry reference compiled from the listed sources and cross-checked against current veterinary guidance at the time of publication. It is not a substitute for a veterinarian who treats fish, and nothing in it should delay a vet visit. Medication is dosed to its label or a vet's direction, never to a number read in a book.

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